CreaseBench: question booklet

34 questions on flat-folded origami: reading crease patterns, folding, tree theory, flap design and defect finding. Each question is exactly the prompt a model receives. The model conventions are printed once below and apply to every question of that tier.

How to answer. Give the JSON answer each question asks for. Answers are graded by actually folding them, so partial sketches score nothing. For large inputs, use the files in `inputs/Qnnn/`; any tool you like is allowed (ruler, compass, paper, a computer) unless you are timing a closed-book attempt.

split	questions	n	contents
expert	Q001–Q034	34	Headline split, graded pass/fail. Expert-level design, image-only tree reading and inverse folding.

Conventions: advanced tier

You are taking CreaseBench, a test of spatial and analytical reasoning about flat-folded origami.

Conventions (all tasks):

- The paper is the unit square [0,1] x [0,1], seen from its top side. Lengths are in units of the paper side.
- Crease patterns (CPs) use the FOLD format: `vertices_coords`, `edges_vertices`, `edges_assignment` ("M" mountain, "V" valley, "B" paper boundary, "U" folded but direction unspecified) and `faces_vertices` (counter-clockwise vertex loops; face indices refer to this list).
- Valley fold: the paper on both sides of the crease rises toward the viewer, so the moving part ends up ON TOP. Mountain fold: the opposite (it ends up underneath).
- A crease pattern describes a flat-folded state in which EVERY crease is folded completely flat (by 180 degrees) and the paper is otherwise unbent. No paper may pass through other paper.
- Folded coordinates: report them as seen from above, without mirroring. Any position and rotation is fine (answers are aligned rigidly before grading), but the scale must be true.
- Layer order: a pair [f, g] in `face_above` means face f lies above face g (closer to the viewer) where they overlap, in the frame of your reported coordinates.
- Give your final answer as a single JSON object in a ```json fenced code block at the end of your reply. Numbers should have at least 6 significant digits.

Tree theory conventions (advanced tasks):

- A flat-folded base is described by its stick-figure TREE: project the folded model onto its axis (the line its flaps lie along) and contract every connected piece of paper at the same axis coordinate to a point. Leaves are flap tips; edge lengths are distances along the axis, in units of the paper side.
- A tree is given as JSON `{"edges": [[node, node, length], ...]}`; node names are arbitrary labels.
- A design realizes a target tree at SCALE m if its flap tips can be matched to the target's leaves so that for every pair of target leaves (i, j) the tree distance between the matched tips is at least m * d_target(i, j) (Lang's path condition: flaps may be shortened or bundled afterwards). The paper is ONE uncut unit square. Bigger m is better; it is compared with the best known circle/river packing.
- Design systems: box pleating on an N-grid = every crease at 0/45/90/135 degrees with vertices on the 1/N lattice; 22.5-degree design = every crease at a multiple of 22.5 degrees; a Pythagorean stretch = a box-pleated pattern that also uses lattice directions from a Pythagorean triple (a, b, c) (slopes b/a, a/(b+c), b/(a+c), ...) to place flaps that are diagonally offset; grafting = inserting strips of paper into an existing crease pattern.

Q001

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q001/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a crab: two claws, eight legs, eyes on stalks). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0957510 (one unit of tree length -> 0.0957510 of the paper side).

Input `tree`:

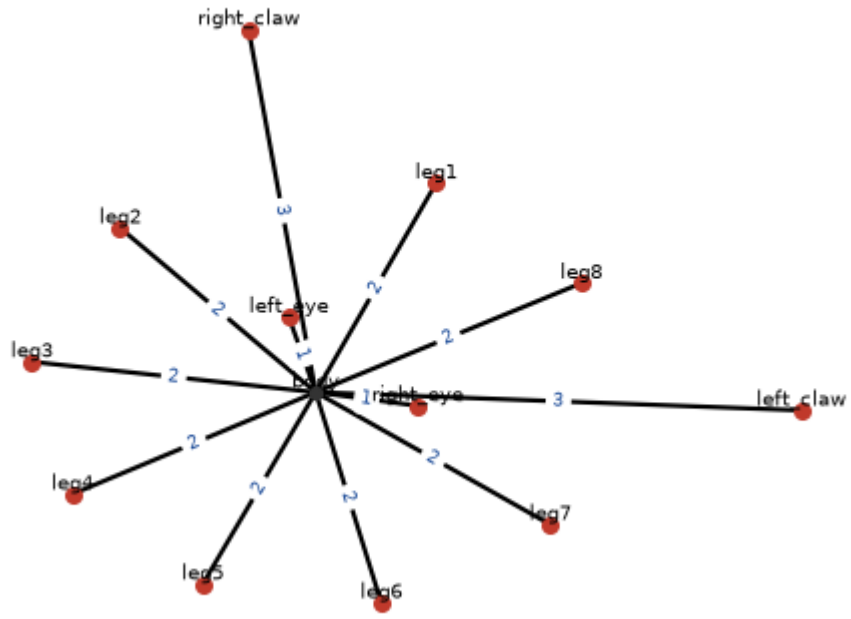

```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",
1],["body","leg1",2],["body","leg2",2],["body","leg3",2],["body","leg4",2],["body","leg5",2],["bod
y","leg6",2],["body","leg7",2],["body","leg8",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q002

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q002/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a dragon: head with two horns, two wings, four legs, long tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0687332 (one unit of tree length -> 0.0687332 of the paper side).

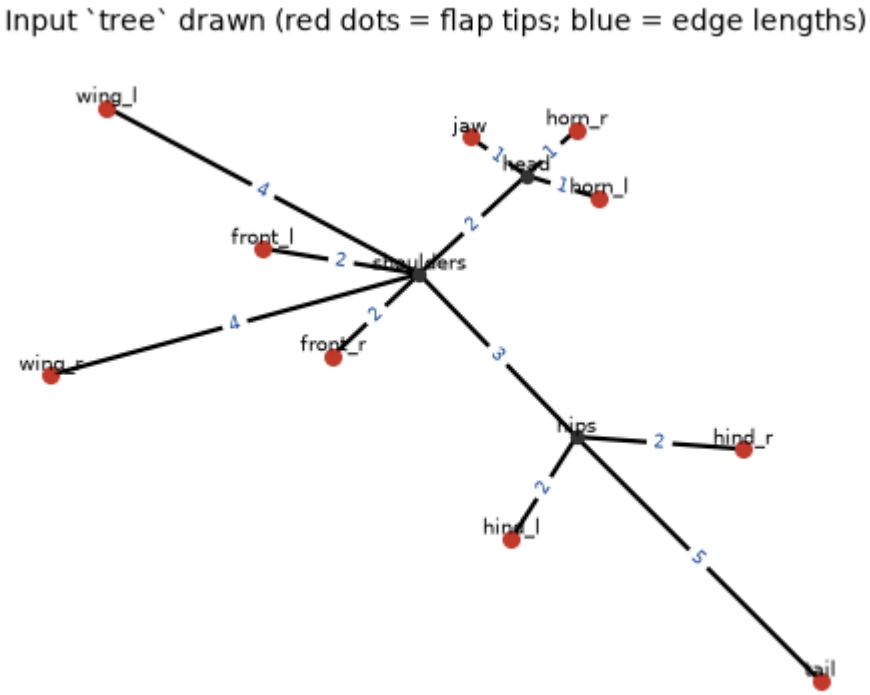
Input `tree`:


```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulder
s","wing_l",4],["shoulders","wing_r",4],["shoulders","front_l",2],["shoulders","front_r",2],["shoulde
rs","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



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Working / answer

Q003

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q003/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a lobster: two big claws, long antennae, antennules, eight walking legs, a three-lobed tail fan). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0453081 (one unit of tree length -> 0.0453081 of the paper side).

Input `tree`:

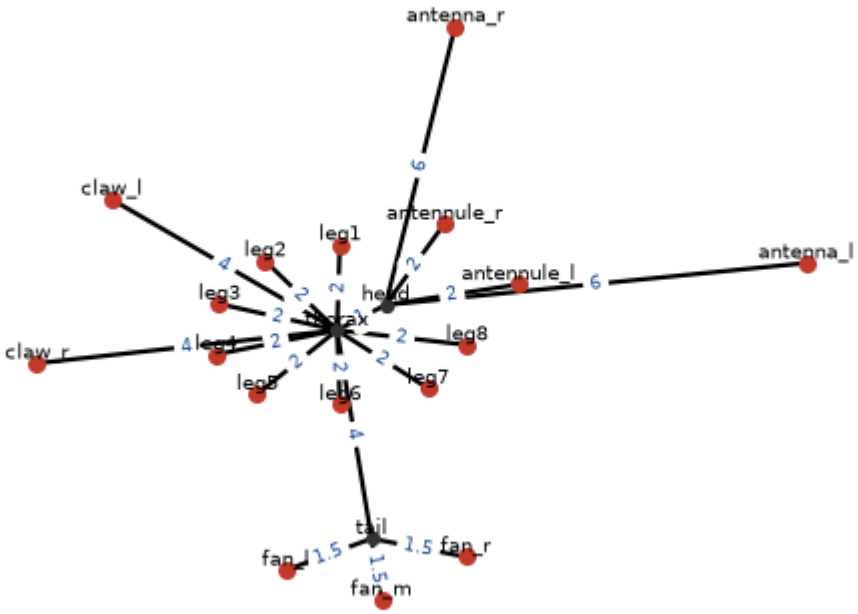

```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_
r",2],["head","thorax",1],["thorax","claw_l",4],["thorax","claw_r",4],["thorax","leg1",2],["thora
x","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thora
x","leg7",2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tai
l","fan_r",1.5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q004

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q004/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a praying mantis: antennae, raptorial forelegs, four walking legs, four wings, abdomen with two cerci). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0443844 (one unit of tree length -> 0.0443844 of the paper side).

Input `tree`:

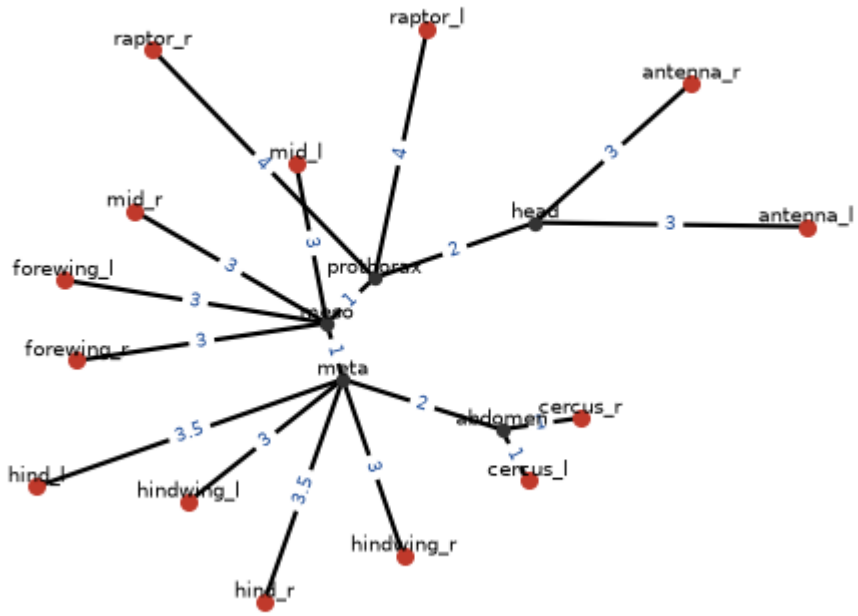

```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_
l",4],["prothorax","raptor_r",4],["prothorax","meso",1],["meso","mid_l",3],["meso","mid_r",3],["mes
o","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",
3.5],["meta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],
["abdomen","cercus_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q005

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q005/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a pegasus: head with two ears and a muzzle, wings with three primary feathers each, four legs, a tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0432756 (one unit of tree length -> 0.0432756 of the paper side).

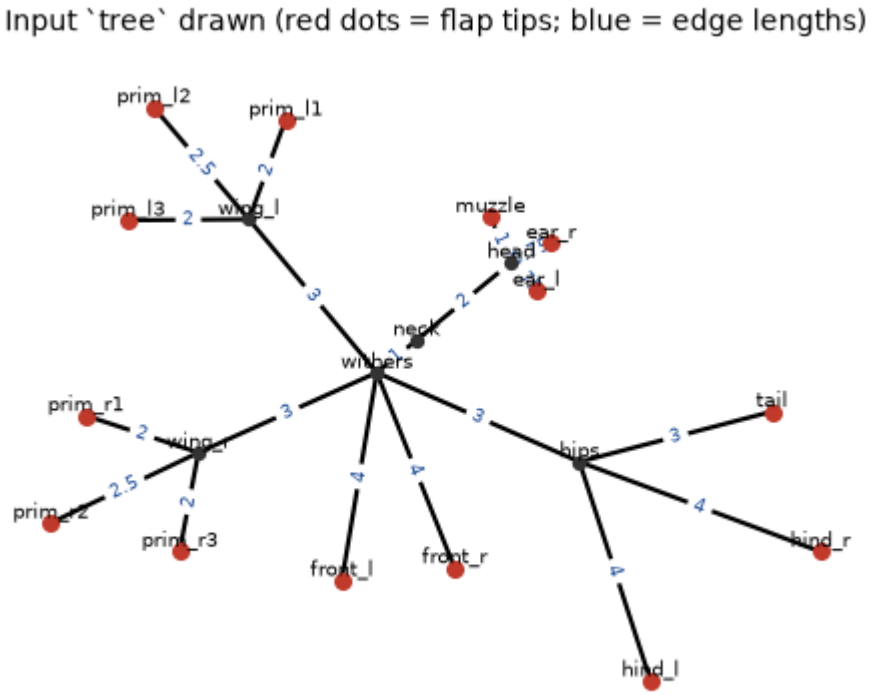
Input `tree`:


```
{"edges":[["neck","head",2],["head","ear_l",0.75],["head","ear_r",0.75],["head","muzzle",1],["nec
k","withers",1],["withers","wing_l",3],["wing_l","prim_l1",2],["wing_l","prim_l2",2.5],["wing_l","pri
m_l3",2],["withers","wing_r",3],["wing_r","prim_r1",2],["wing_r","prim_r2",2.5],["wing_r","prim_r3",
2],["withers","front_l",4],["withers","front_r",4],["withers","hips",3],["hips","hind_l",4],["hip
s","hind_r",4],["hips","tail",3]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



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Working / answer

Q006

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q006/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a wingless wyrm: a head with two horns, two forelegs, and twelve dorsal scales along a long body ending in a tail fin). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0609262 (one unit of tree length -> 0.0609262 of the paper side).

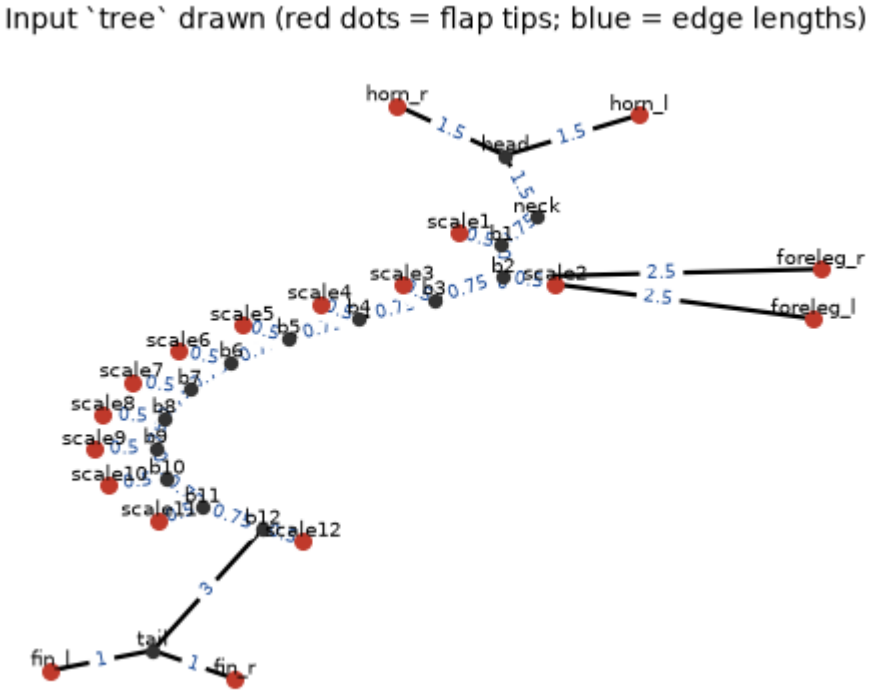
Input `tree`:


```
{"edges":[["neck","head",1.5],["head","horn_l",1.5],["head","horn_r",1.5],["neck","b1",0.75],["b1","b
2",0.75],["b2","b3",0.75],["b3","b4",0.75],["b4","b5",0.75],["b5","b6",0.75],["b6","b7",0.75],["b
7","b8",0.75],["b8","b9",0.75],["b9","b10",0.75],["b10","b11",0.75],["b11","b12",0.75],["b1","scale
1",0.5],["b2","scale2",0.5],["b3","scale3",0.5],["b4","scale4",0.5],["b5","scale5",0.5],["b6","scale
6",0.5],["b7","scale7",0.5],["b8","scale8",0.5],["b9","scale9",0.5],["b10","scale10",0.5],["b11","sca
le11",0.5],["b12","scale12",0.5],["b2","foreleg_l",2.5],["b2","foreleg_r",2.5],["b12","tail",3],["tai
l","fin_l",1],["tail","fin_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



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Working / answer

Q007

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q007/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a stag beetle: long mandibles, antennae, palps, six legs, two elytra (wings), abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0454431 (one unit of tree length -> 0.0454431 of the paper side).

Input `tree`:

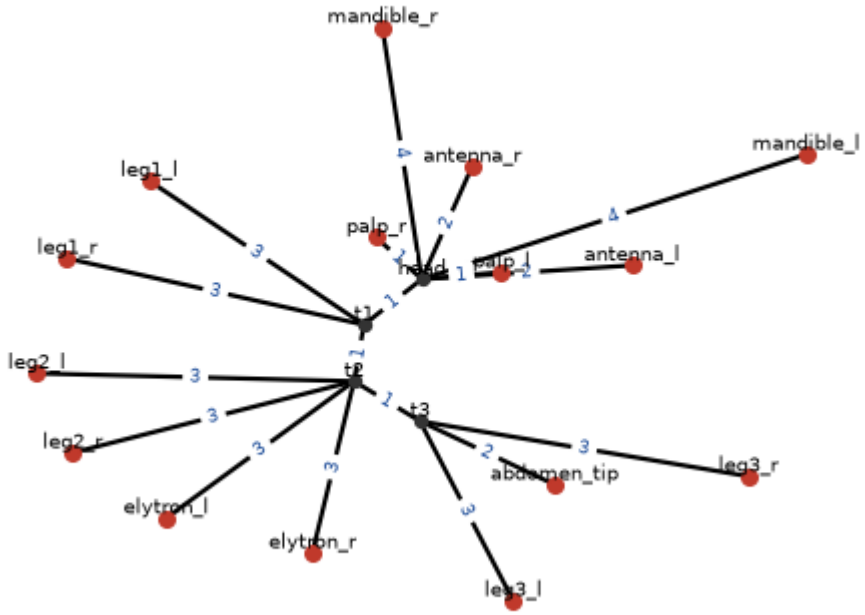

```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",
2],["head","palp_l",1],["head","palp_r",1],["head","t1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t
1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",
1],["t3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q008

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q008/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a tarantula: eight long legs, pedipalps, chelicerae, spinnerets on a big abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0516059 (one unit of tree length -> 0.0516059 of the paper side).

Input `tree`:

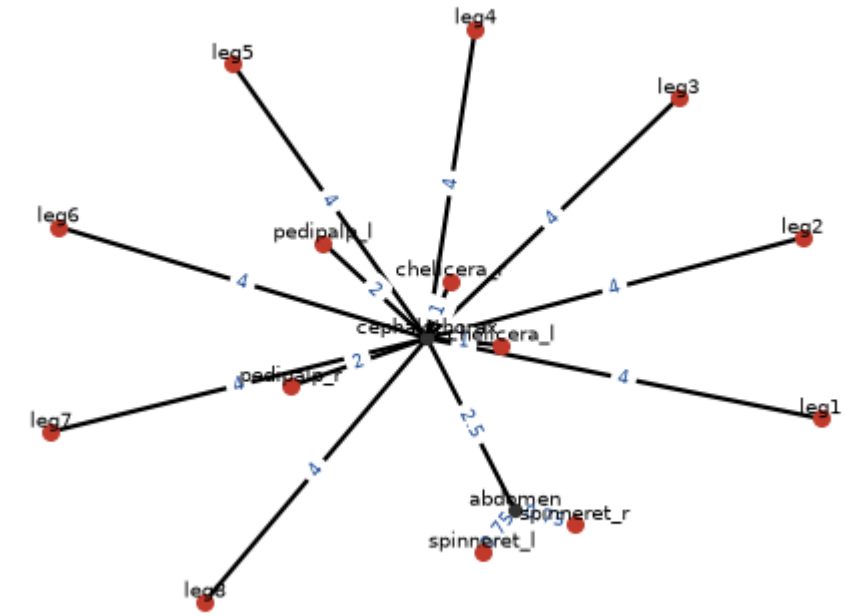

```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalothorax","leg4",4],["cephalothorax","leg5",4],["cephalothorax","leg6",4],["cephalothorax","leg7",4],["cephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephalothorax","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdomen","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q009

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q009/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a crab: two claws, eight legs, eyes on stalks). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0957510 (one unit of tree length -> 0.0957510 of the paper side). In addition, no point of the folded base may have more than 74 layers of paper.

Input `tree`:

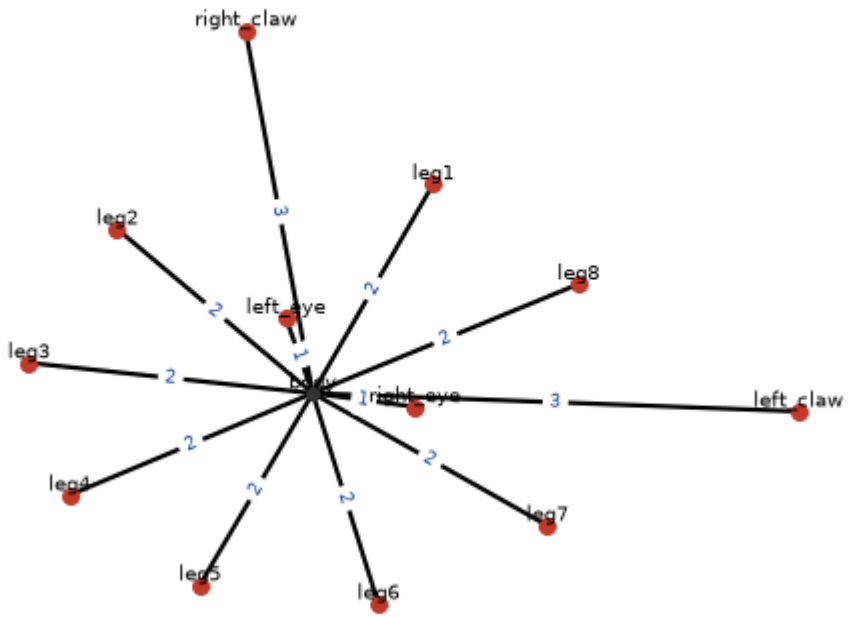

```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",
1],["body","leg1",2],["body","leg2",2],["body","leg3",2],["body","leg4",2],["body","leg5",2],["bod
y","leg6",2],["body","leg7",2],["body","leg8",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q010

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q010/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a dragon: head with two horns, two wings, four legs, long tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0687332 (one unit of tree length -> 0.0687332 of the paper side). In addition, no point of the folded base may have more than 36 layers of paper.

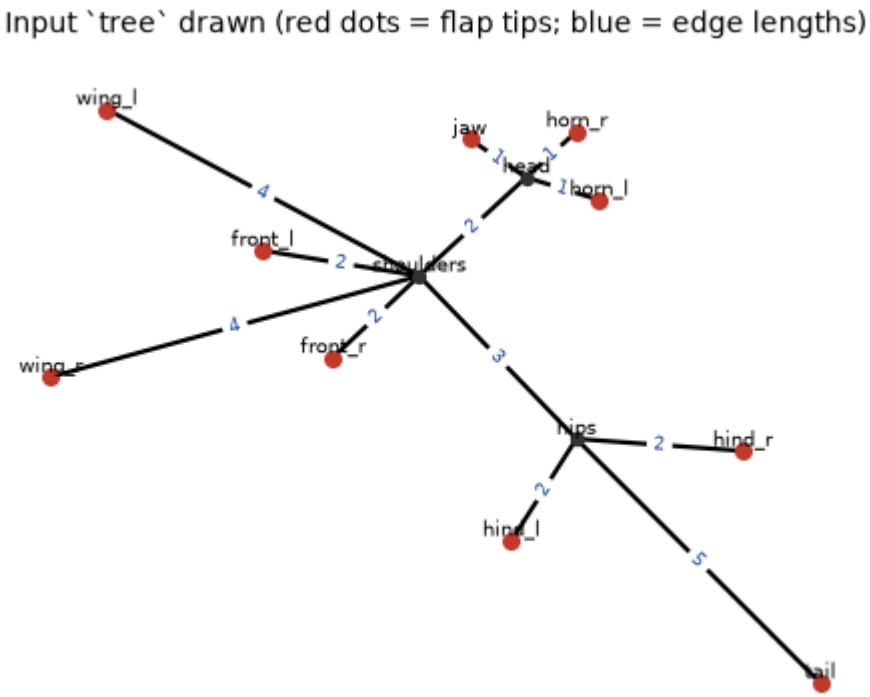
Input `tree`:


```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulder
s","wing_l",4],["shoulders","wing_r",4],["shoulders","front_l",2],["shoulders","front_r",2],["shoulde
rs","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q011

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q011/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a lobster: two big claws, long antennae, antennules, eight walking legs, a three-lobed tail fan). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0453081 (one unit of tree length -> 0.0453081 of the paper side). In addition, no point of the folded base may have more than 100 layers of paper.

Input `tree`:

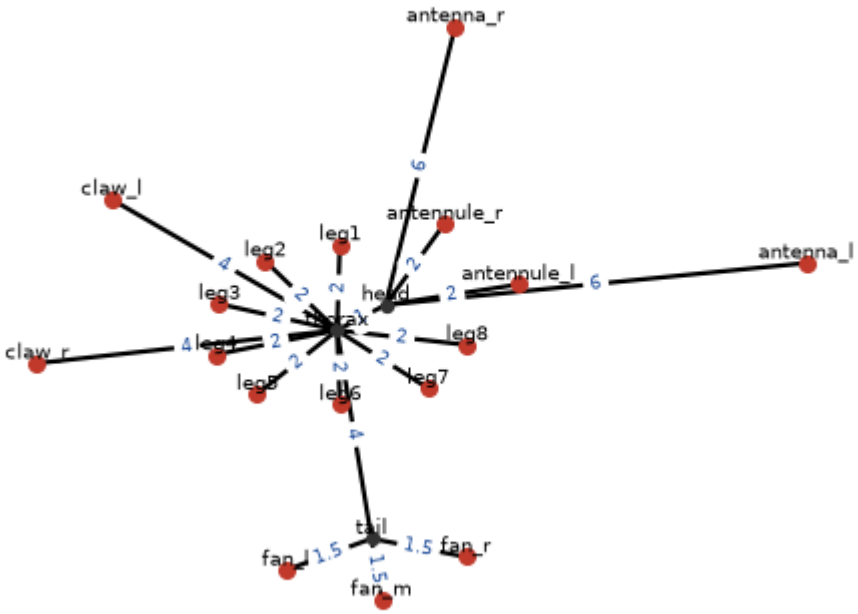

```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_
r",2],["head","thorax",1],["thorax","claw_l",4],["thorax","claw_r",4],["thorax","leg1",2],["thora
x","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thora
x","leg7",2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tai
l","fan_r",1.5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q012

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q012/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a praying mantis: antennae, raptorial forelegs, four walking legs, four wings, abdomen with two cerci). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0443844 (one unit of tree length -> 0.0443844 of the paper side). In addition, no point of the folded base may have more than 60 layers of paper.

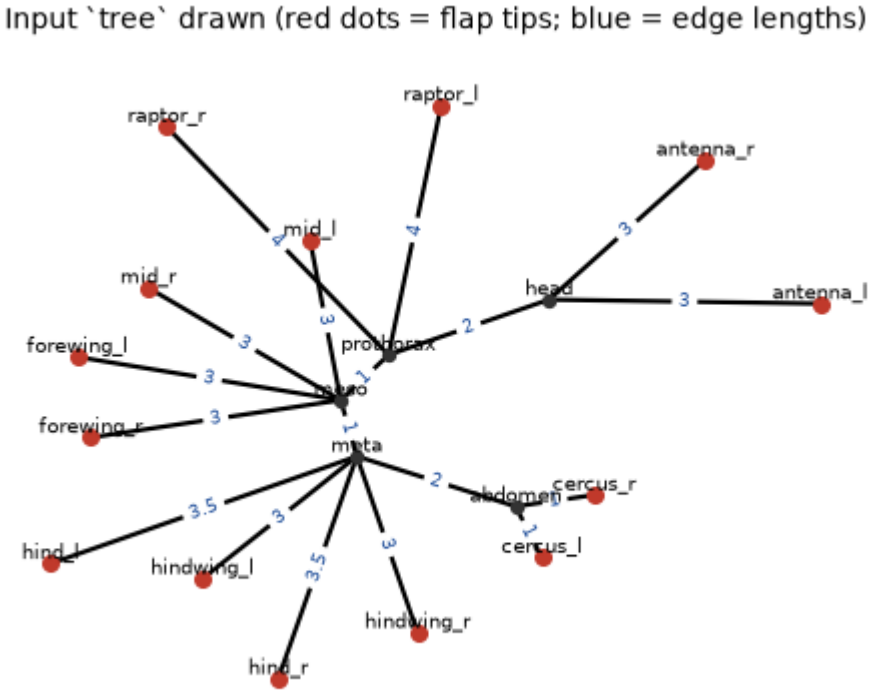
Input `tree`:


```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_
l",4],["prothorax","raptor_r",4],["prothorax","meso",1],["meso","mid_l",3],["meso","mid_r",3],["mes
o","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",
3.5],["meta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],
["abdomen","cercus_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q013

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q013/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a pegasus: head with two ears and a muzzle, wings with three primary feathers each, four legs, a tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0432756 (one unit of tree length -> 0.0432756 of the paper side). In addition, no point of the folded base may have more than 62 layers of paper.

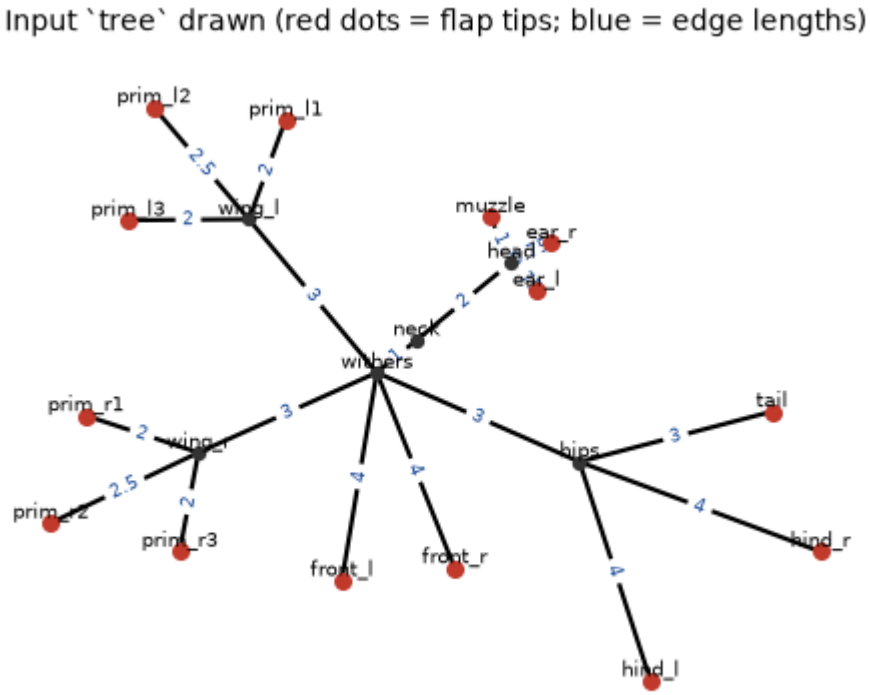
Input `tree`:


```
{"edges":[["neck","head",2],["head","ear_l",0.75],["head","ear_r",0.75],["head","muzzle",1],["nec
k","withers",1],["withers","wing_l",3],["wing_l","prim_l1",2],["wing_l","prim_l2",2.5],["wing_l","pri
m_l3",2],["withers","wing_r",3],["wing_r","prim_r1",2],["wing_r","prim_r2",2.5],["wing_r","prim_r3",
2],["withers","front_l",4],["withers","front_r",4],["withers","hips",3],["hips","hind_l",4],["hip
s","hind_r",4],["hips","tail",3]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q014

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q014/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a wingless wyrm: a head with two horns, two forelegs, and twelve dorsal scales along a long body ending in a tail fin). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0609262 (one unit of tree length -> 0.0609262 of the paper side). In addition, no point of the folded base may have more than 54 layers of paper.

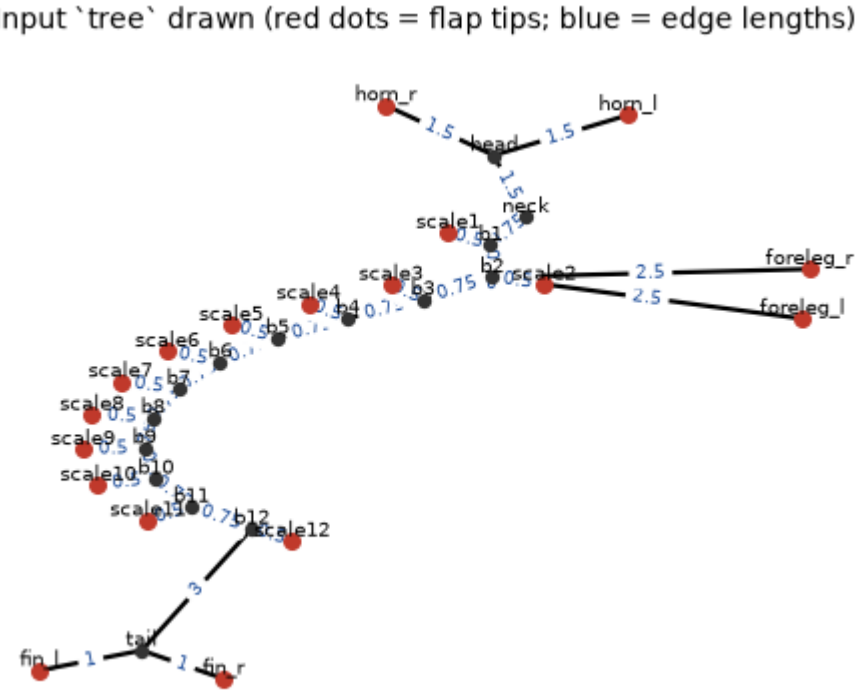
Input `tree`:


```
{"edges":[["neck","head",1.5],["head","horn_l",1.5],["head","horn_r",1.5],["neck","b1",0.75],["b1","b
2",0.75],["b2","b3",0.75],["b3","b4",0.75],["b4","b5",0.75],["b5","b6",0.75],["b6","b7",0.75],["b
7","b8",0.75],["b8","b9",0.75],["b9","b10",0.75],["b10","b11",0.75],["b11","b12",0.75],["b1","scale
1",0.5],["b2","scale2",0.5],["b3","scale3",0.5],["b4","scale4",0.5],["b5","scale5",0.5],["b6","scale
6",0.5],["b7","scale7",0.5],["b8","scale8",0.5],["b9","scale9",0.5],["b10","scale10",0.5],["b11","sca
le11",0.5],["b12","scale12",0.5],["b2","foreleg_l",2.5],["b2","foreleg_r",2.5],["b12","tail",3],["tai
l","fin_l",1],["tail","fin_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q015

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q015/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a stag beetle: long mandibles, antennae, palps, six legs, two elytra (wings), abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0454431 (one unit of tree length -> 0.0454431 of the paper side). In addition, no point of the folded base may have more than 62 layers of paper.

Input `tree`:

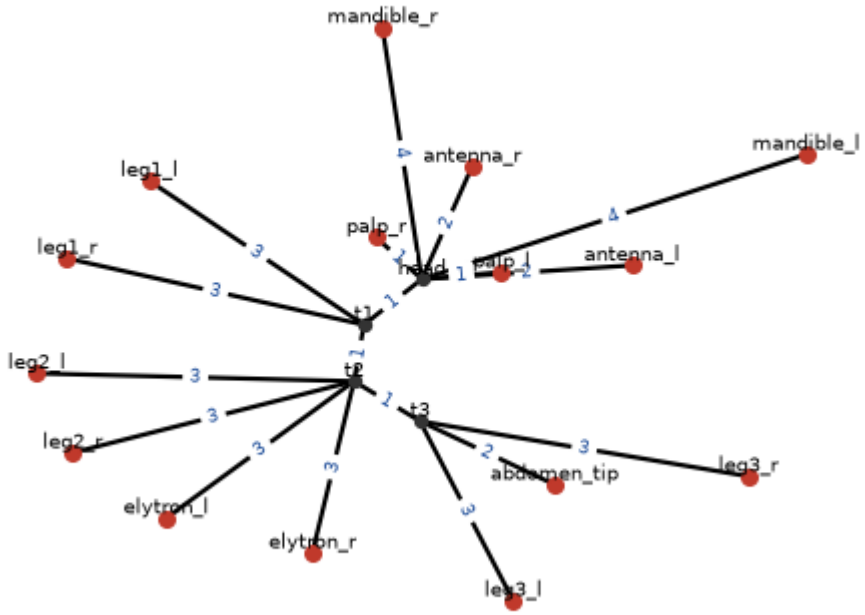

```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",
2],["head","palp_l",1],["head","palp_r",1],["head","t1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t
1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",
1],["t3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q016

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q016/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a tarantula: eight long legs, pedipalps, chelicerae, spinnerets on a big abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0516059 (one unit of tree length -> 0.0516059 of the paper side). In addition, no point of the folded base may have more than 76 layers of paper.

Input `tree`:

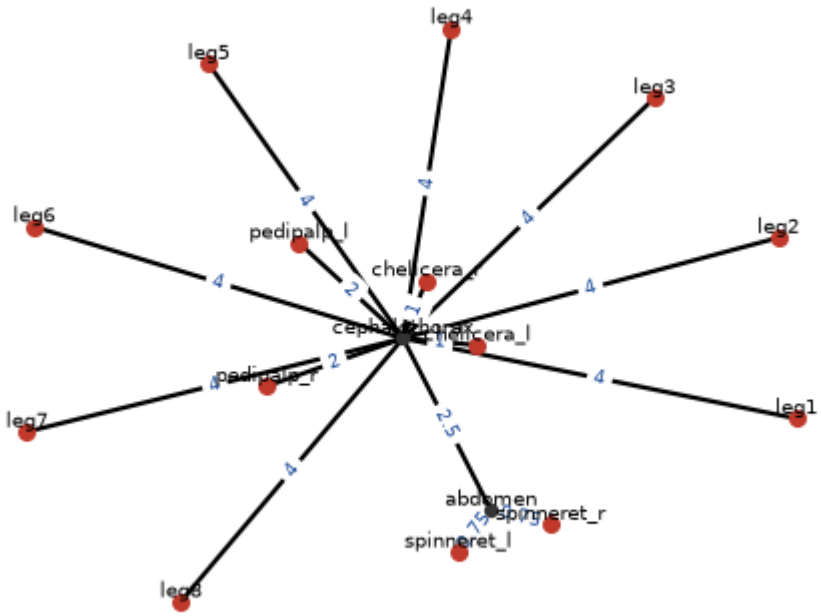

```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalothorax","leg4",4],["cephalothorax","leg5",4],["cephalothorax","leg6",4],["cephalothorax","leg7",4],["cephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephalothorax","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdomen","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q017

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q017/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.09384, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",
1],["body","leg1",2],["body","leg2",2],["body","leg3",2],["body","leg4",2],["body","leg5",2],["bod
y","leg6",2],["body","leg7",2],["body","leg8",2]]}
```

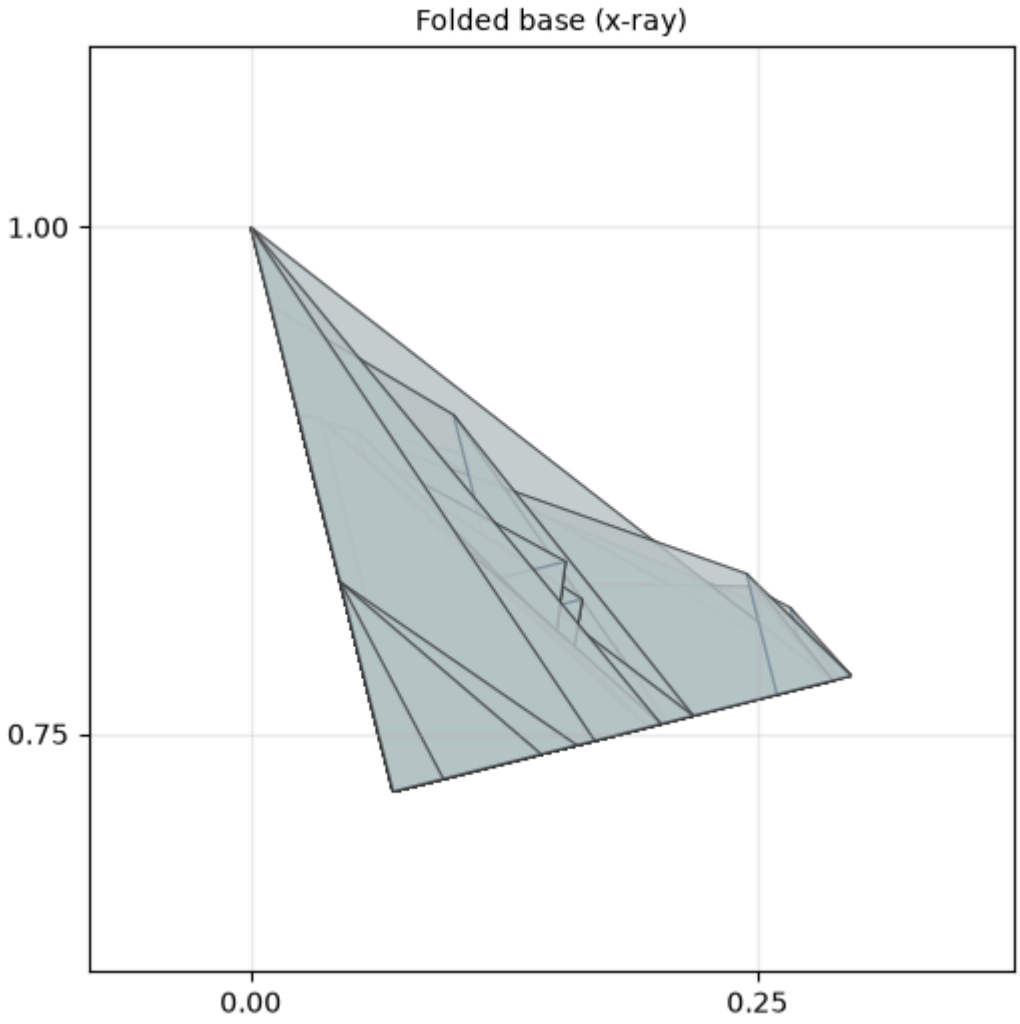

Input `folded_outline`:


```
[[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],
[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],
[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],
[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],
[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],
[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],
[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[1e-06,1.0],[1e-06,1.0],[1e-06,1.0],[1e-0
6,1.0],[0.198822,0.84513],[0.198822,0.84513],[0.198822,0.84513],[0.198822,0.84513],[0.200861,0.84440
9],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.82892
2],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.25669,0.81718],
[0.25669,0.81718],[0.25669,0.81718],[0.25669,0.81718],[0.26625,0.812132],[0.26625,0.812132],[0.26 ...
[truncated: the full input is in inputs/Q017/]
```

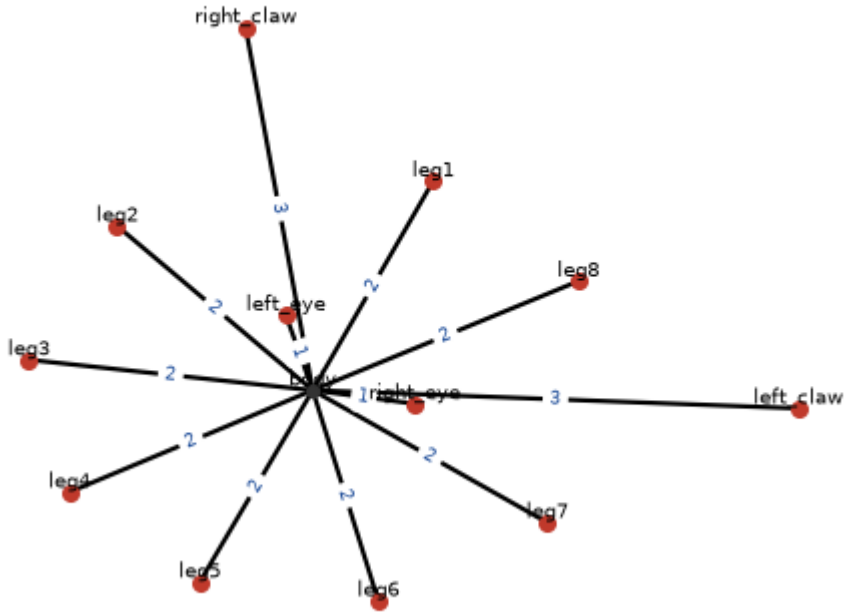

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q018

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q018/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.06736, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulder
s","wing_l",4],["shoulders","wing_r",4],["shoulders","front_l",2],["shoulders","front_r",2],["shoulde
rs","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```

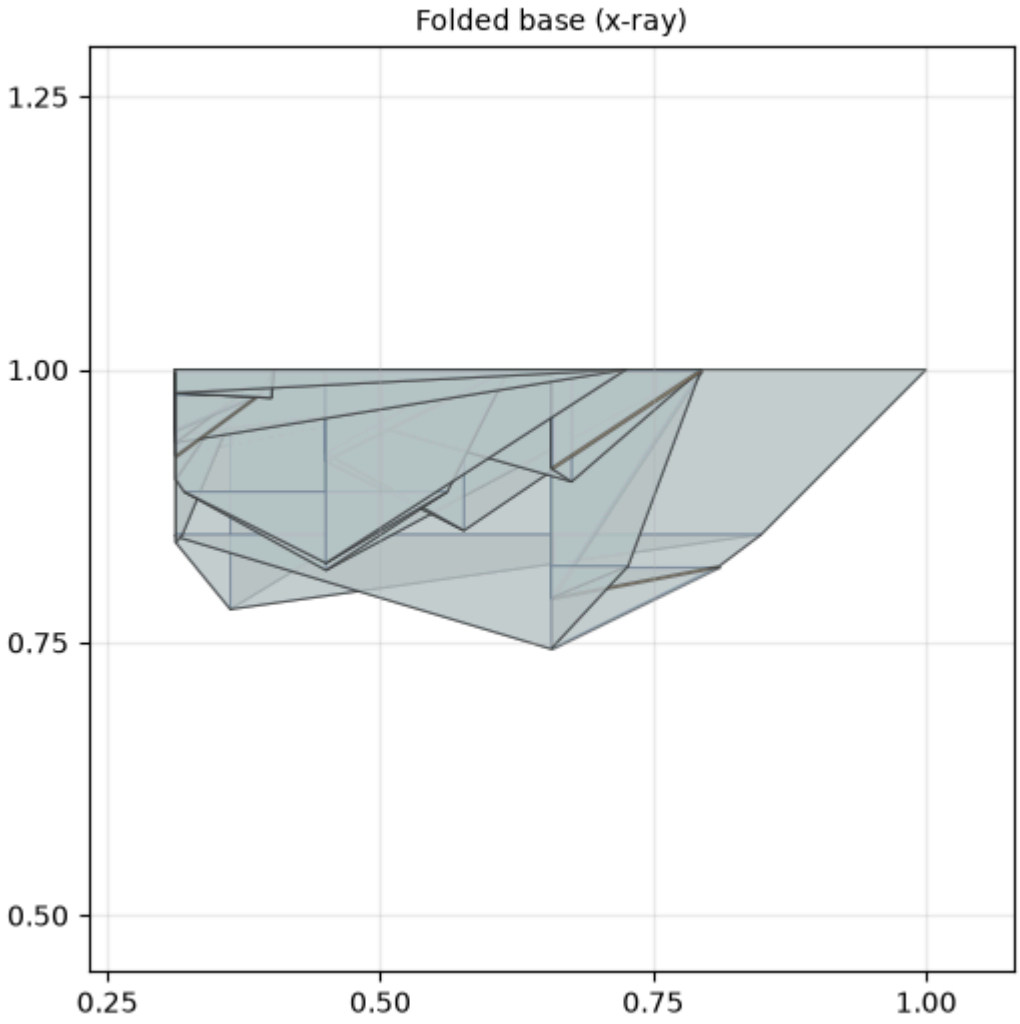

Input `folded_outline`:


```
[[0.312664,1.0],[0.312664,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,
1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312
665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],
[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.32145,1.0],[0.362533,1.0],[0.362533,1.
0],[0.362533,1.0],[0.362533,1.0],[0.362533,1.0],[0.368039,1.0],[0.450132,1.0],[0.673527,1.0],[0.79379
9,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.7
93799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],
[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,
1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793
799,1.0],[0.793799,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.
0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[1.0,1.
0],[1.0,1.0],[1.0,1.0],[0.848854,0.848854],[0.848854,0.848854],[0.848854,0.848854],[0.848854,0 ...
[truncated: the full input is in inputs/Q018/]
```

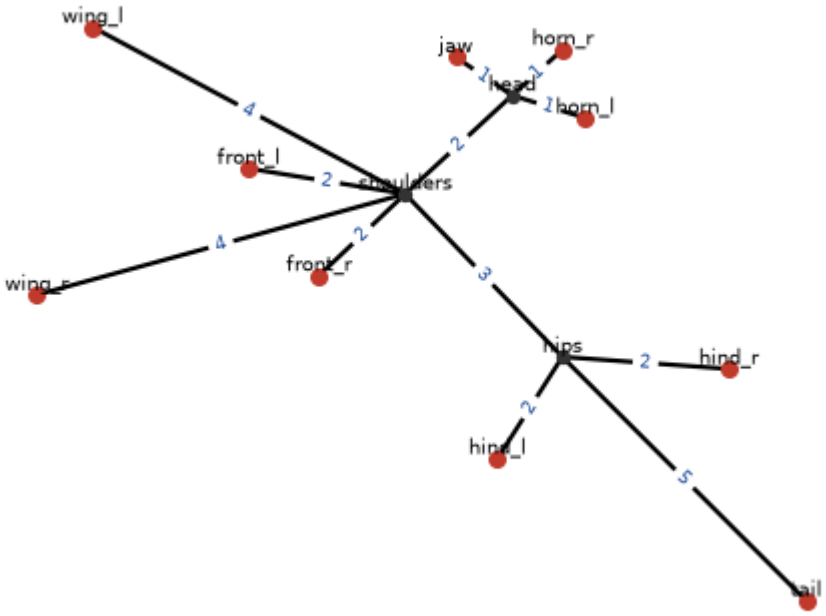

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q019

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q019/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04440, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_
r",2],["head","thorax",1],["thorax","claw_l",4],["thorax","claw_r",4],["thorax","leg1",2],["thora
x","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thora
x","leg7",2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tai
l","fan_r",1.5]]}
```

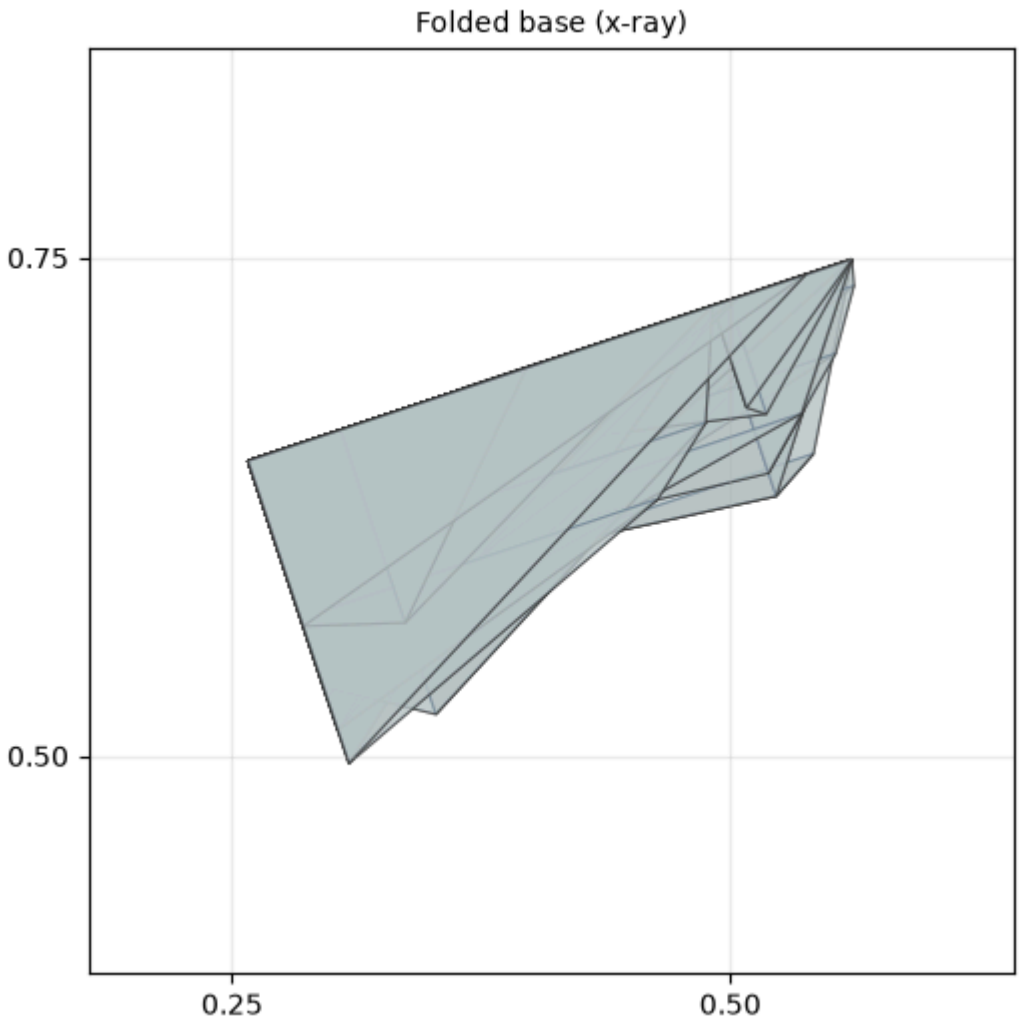

Input `folded_outline`:


```
[[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],
[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],
[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],
[0.257938,0.648069],[0.343705,0.676699],[0.504243,0.730289],[0.504243,0.730289],[0.504243,0.730289],
[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],
[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],
[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],
[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],
[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288],
[0.504243,0.730289],[0.504243,0.730289],[0.504243,0.730289],[0.504239,0.730286],[0.504239,0.730286],
[0.504239,0.730286],[0.504239,0.730286],[0.504239,0.730286],[0.504239,0.730286],[0.504239,0.730286]
...
[truncated: the full input is in inputs/Q019/]
```

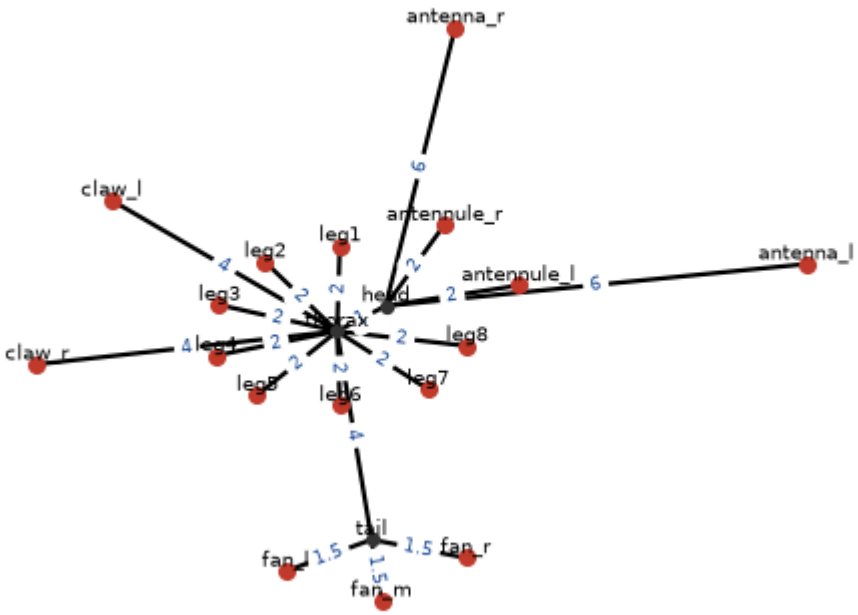

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q020

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q020/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04350, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_
l",4],["prothorax","raptor_r",4],["prothorax","meso",1],["meso","mid_l",3],["meso","mid_r",3],["mes
o","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",
3.5],["meta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],
["abdomen","cercus_r",1]]}
```

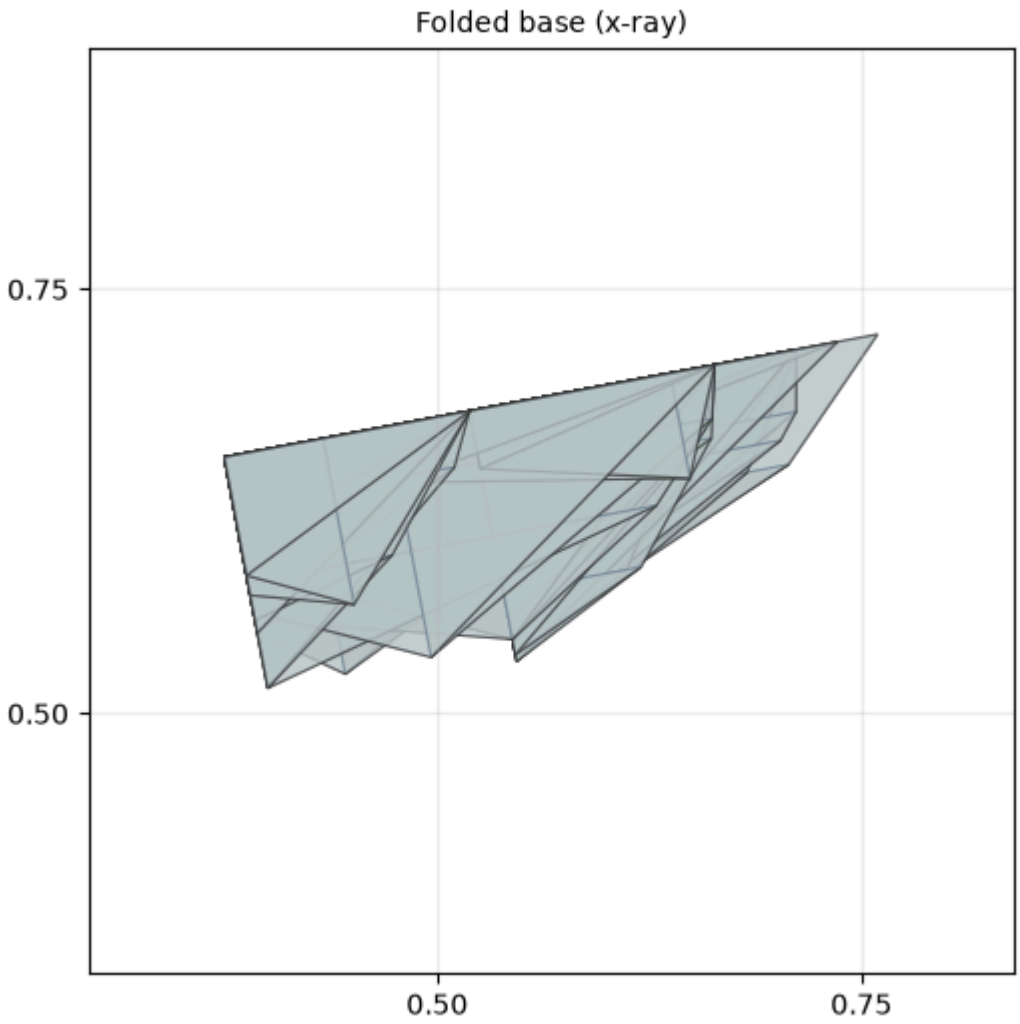

Input `folded_outline`:


```
[[[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.65102
1],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.65102
1],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.65102
1],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.65102
1],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.65102
1],[0.374575,0.651021],[0.374776,0.651058],[0.519028,0.678177],[0.519028,0.678177],[0.519028,0.67817
7],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.67817
7],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.67817
7],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.67817
7],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.67817
7],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.67817
7],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519028,0.678177
...
[truncated: the full input is in inputs/Q020/]
```

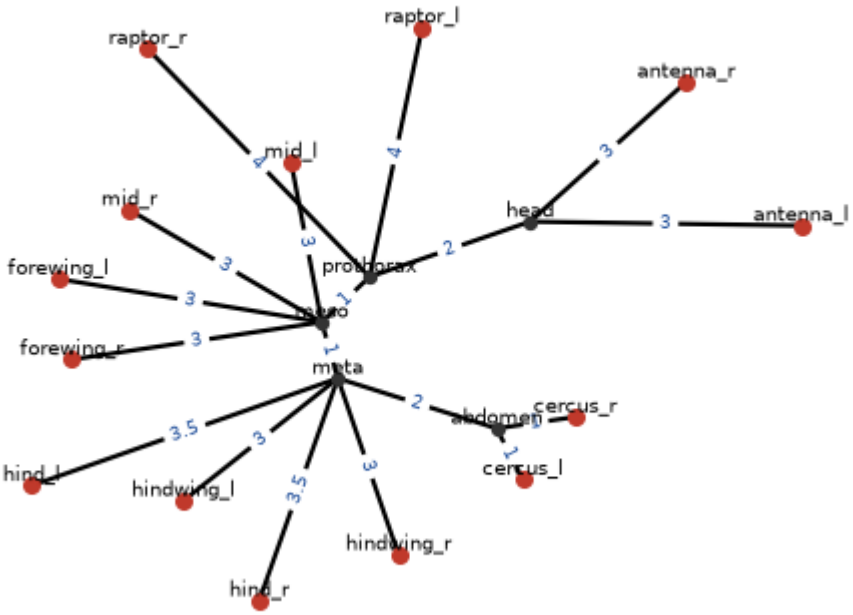

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q021

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q021/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04241, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["neck","head",2],["head","ear_l",0.75],["head","ear_r",0.75],["head","muzzle",1],["nec
k","withers",1],["withers","wing_l",3],["wing_l","prim_l1",2],["wing_l","prim_l2",2.5],["wing_l","pri
m_l3",2],["withers","wing_r",3],["wing_r","prim_r1",2],["wing_r","prim_r2",2.5],["wing_r","prim_r3",
2],["withers","front_l",4],["withers","front_r",4],["withers","hips",3],["hips","hind_l",4],["hip
s","hind_r",4],["hips","tail",3]]}
```

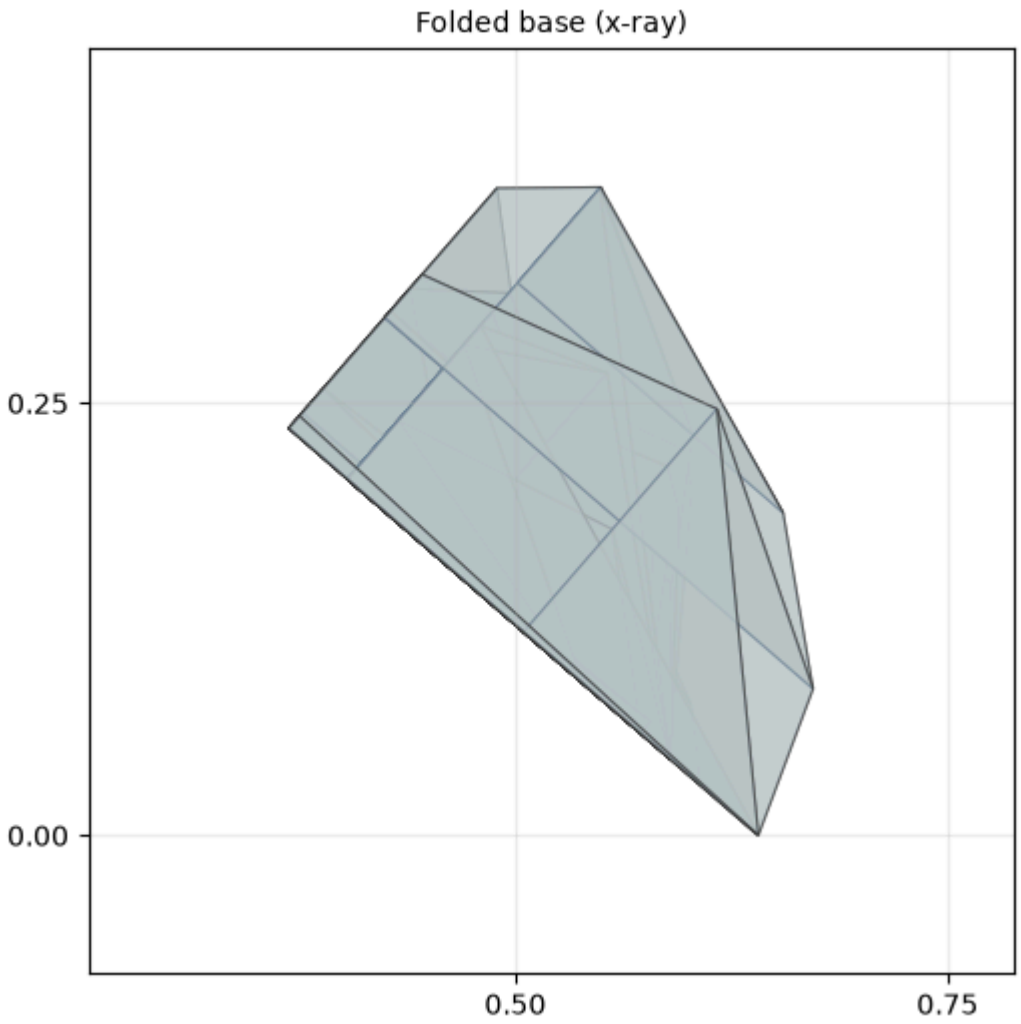

Input `folded_outline`:


```
[[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],
[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369036,0.235491],[0.369036,0.235491],
[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],
[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],
[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],
[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],
[0.369036,0.235491],[0.369036,0.235491],[0.370136,0.23676],[0.376846,0.244497],[0.376846,0.244497],
[0.376846,0.244497],[0.376846,0.244497],[0.376846,0.244497],[0.388902,0.258398],[0.388902,0.258398],
[0.388902,0.258398],[0.388902,0.258398],[0.388902,0.258398],[0.388902,0.258398],[0.388902,0.258398],
[0.424627,0.299591],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],
[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],
[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],
...
[truncated: the full input is in inputs/Q021/]
```

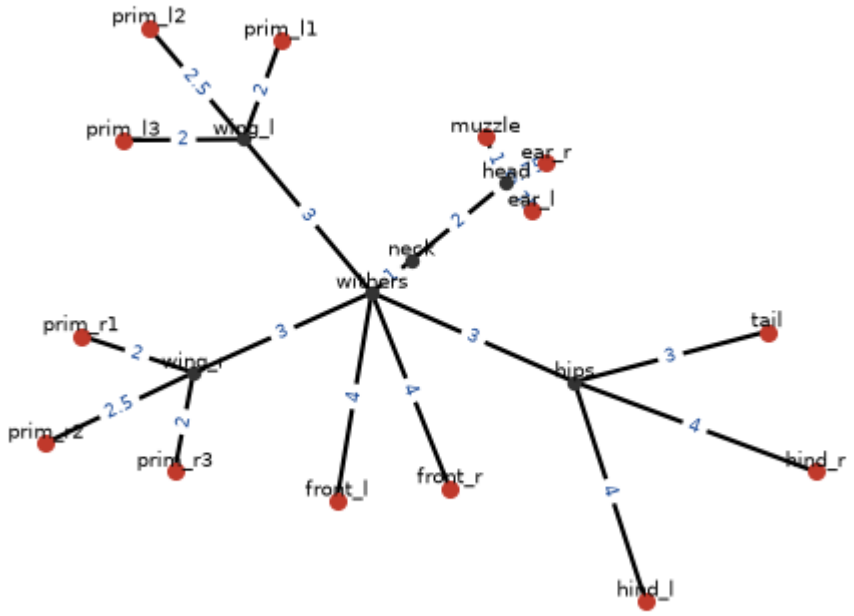

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q022

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q022/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.01470, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["neck","head",1.5],["head","beak",1],["head","crest1",1.5],["head","crest2",2],["head","cr
est3",1.5],["neck","body",1],["body","wing_l",3],["body","wing_r",3],["wing_l","prim_l1",3],["wing_
l","prim_l2",3.5],["wing_l","prim_l3",4],["wing_l","prim_l4",3.5],["wing_l","prim_l5",3],["wing_r","p
rim_r1",3],["wing_r","prim_r2",3.5],["wing_r","prim_r3",4],["wing_r","prim_r4",3.5],["wing_r","prim_r
5",3],["body","leg_l",2],["body","leg_r",2],["leg_l","toe_l1",0.75],["leg_l","toe_l2",0.75],["leg_
l","toe_l3",0.75],["leg_r","toe_r1",0.75],["leg_r","toe_r2",0.75],["leg_r","toe_r3",0.75],["body","ru
mp",1],["rump","tail1",5],["rump","tail2",6],["rump","tail3",7],["rump","tail4",8],["rump","tail5",
7],["rump","tail6",6],["rump","tail7",5]]}
```

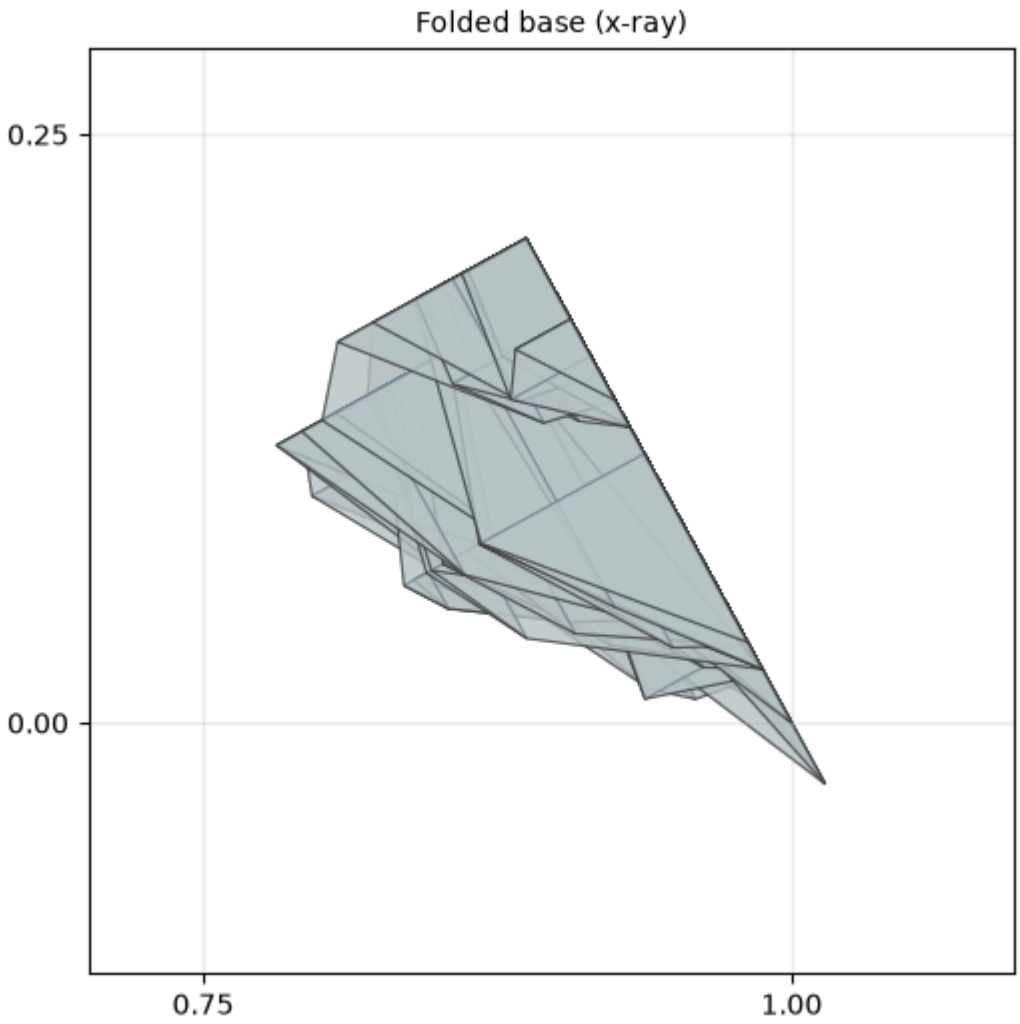

Input `folded_outline`:


```
[[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],
[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],
[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],
[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],
[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],
[0.781281,0.117908],[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],
[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],
[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],
[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909],[0.781283,0.11791],[0.781283,0.11791],[0.
781283,0.11791],[0.781283,0.11791],[0.781283,0.11791],[0.781283,0.11791],[0.781283,0.11791],[0.78128
3,0.11791],[0.781283,0.11791],[0.781283,0.11791],[0.781283,0.11791],[0.781283,0.11791],[0.781283,0
...
[truncated: the full input is in inputs/Q022/]
```

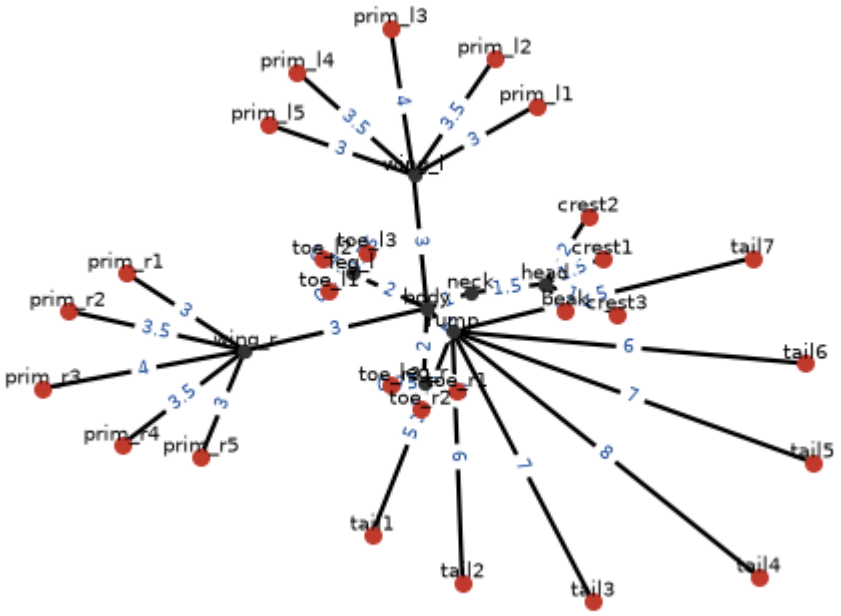

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q023

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q023/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.05971, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["neck","head",1.5],["head","horn_l",1.5],["head","horn_r",1.5],["neck","b1",0.75],["b1","b
2",0.75],["b2","b3",0.75],["b3","b4",0.75],["b4","b5",0.75],["b5","b6",0.75],["b6","b7",0.75],["b
7","b8",0.75],["b8","b9",0.75],["b9","b10",0.75],["b10","b11",0.75],["b11","b12",0.75],["b1","scale
1",0.5],["b2","scale2",0.5],["b3","scale3",0.5],["b4","scale4",0.5],["b5","scale5",0.5],["b6","scale
6",0.5],["b7","scale7",0.5],["b8","scale8",0.5],["b9","scale9",0.5],["b10","scale10",0.5],["b11","sca
le11",0.5],["b12","scale12",0.5],["b2","foreleg_l",2.5],["b2","foreleg_r",2.5],["b12","tail",3],["tai
l","fin_l",1],["tail","fin_r",1]]}
```

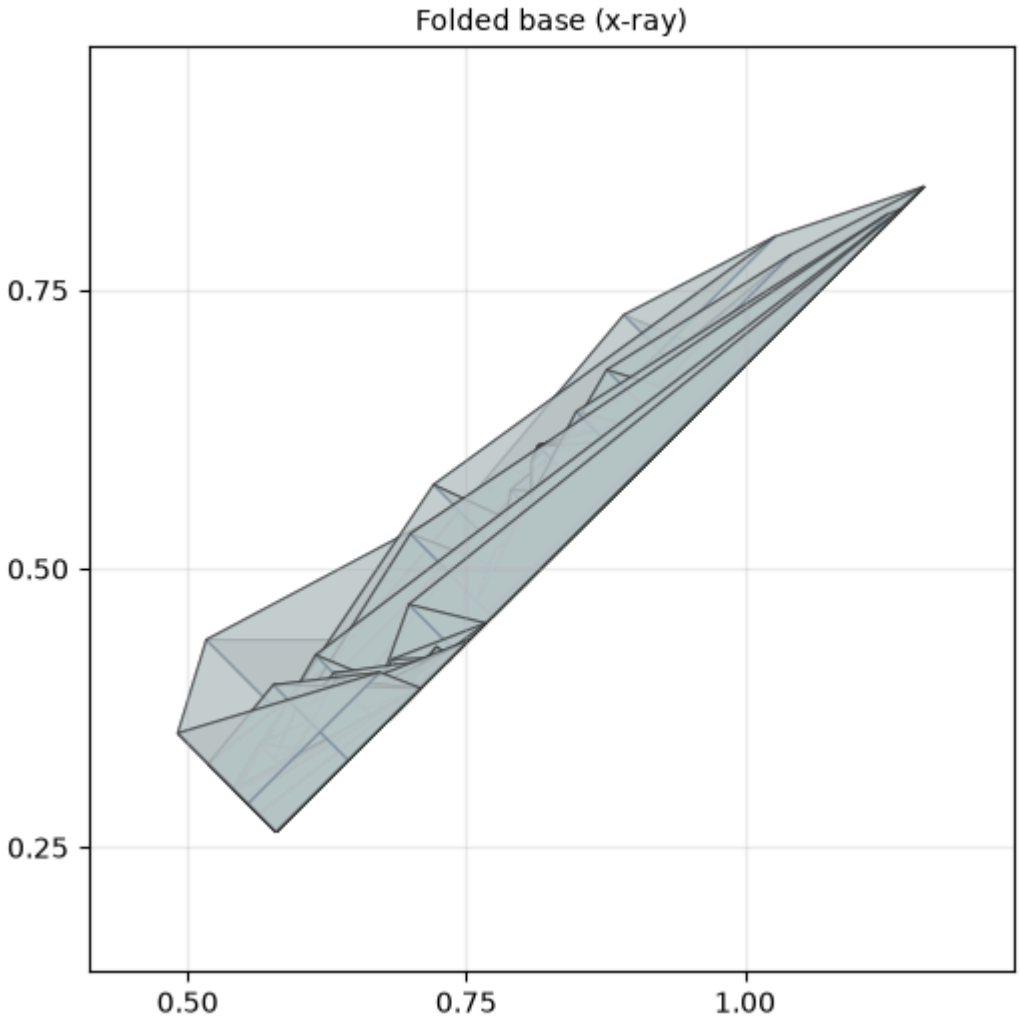

Input `folded_outline`:


```
[[0.491096,0.352392],[0.491096,0.352392],[0.491096,0.352392],[0.491096,0.352392],[0.506259,0.401948],
[0.506259,0.401948],[0.506259,0.401948],[0.516767,0.436292],[0.516767,0.436292],[0.516767,0.436292],
[0.529715,0.443173],[0.5719,0.465591],[0.614085,0.48801],[0.656269,0.510428],[0.688812,0.527722],[0.6
88812,0.527722],[0.688812,0.527722],[0.688812,0.527722],[0.688812,0.527722],[0.688812,0.527722],[0.69
4732,0.536584],[0.720632,0.575356],[0.720632,0.575356],[0.720632,0.575356],[0.722699,0.576857],[0.758
062,0.60255],[0.795491,0.629743],[0.830138,0.654915],[0.830138,0.654915],[0.830138,0.654915],[0.83233
3,0.657526],[0.861865,0.692651],[0.861865,0.692651],[0.879428,0.71354],[0.891397,0.727776],[0.891397,
0.727776],[0.891397,0.727776],[0.891397,0.727776],[0.891397,0.727776],[0.891399,0.727777],[1.027361,
0.798203],[1.027361,0.798203],[1.027361,0.798203],[1.027363,0.798204],[1.027363,0.798204],[1.027363,
0.798204],[1.066317,0.811241],[1.160926,0.842904],[1.160926,0.842904],[1.160926,0.842904],[1.160925,
0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,
0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.1 ...
[truncated: the full input is in inputs/Q023/]
```

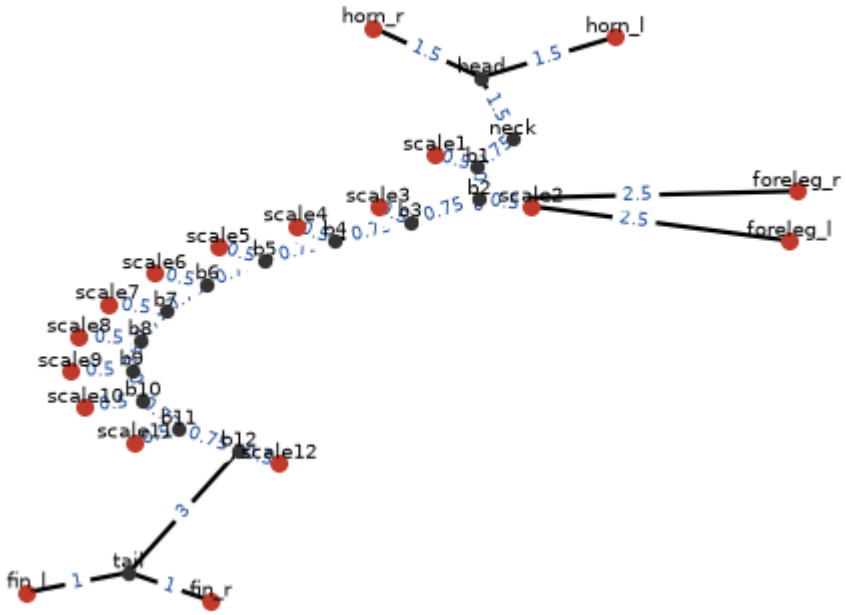

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q024

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q024/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04453, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",
2],["head","palp_l",1],["head","palp_r",1],["head","t1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t
1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",
1],["t3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```

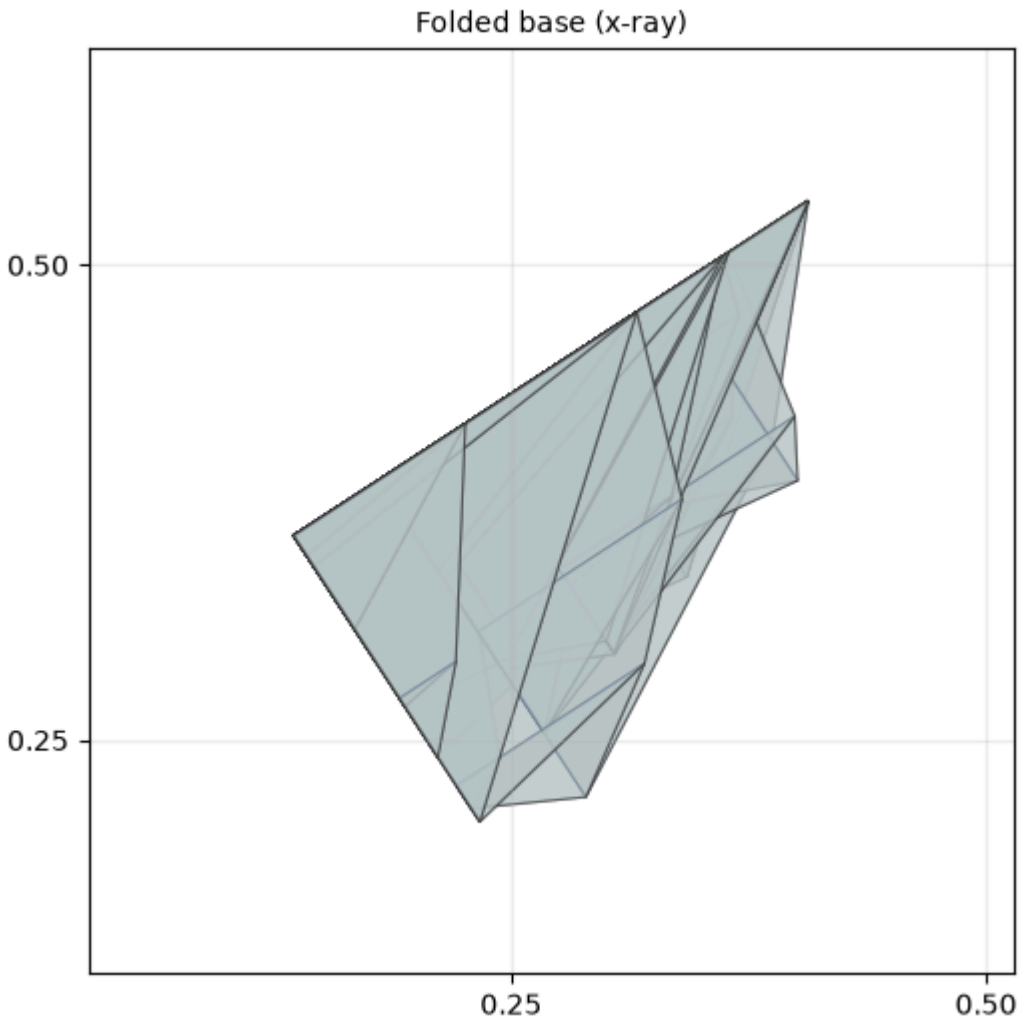

Input `folded_outline`:


```
[[0.135546,0.357646],[0.135546,0.357646],[0.196486,0.397289],[0.316119,0.475114],[0.316119,0.475114],
[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],
[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],
[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],
[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],
[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],
[0.316119,0.475114],[0.316119,0.475114],[0.406405,0.533848],[0.406405,0.533848],[0.406405,0.533848],
[0.406405,0.533848],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],
[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],
[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],
[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],
[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847]
...
[truncated: the full input is in inputs/Q024/]
```

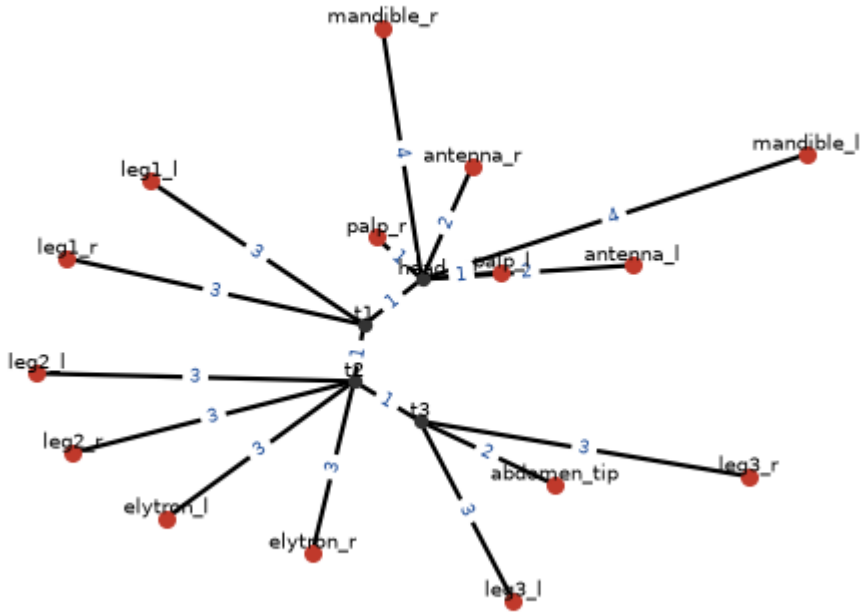

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q025

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q025/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.05057, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalot
horax","leg4",4],["cephalothorax","leg5",4],["cephalothorax","leg6",4],["cephalothorax","leg7",4],["c
ephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephaloth
orax","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdome
n","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```

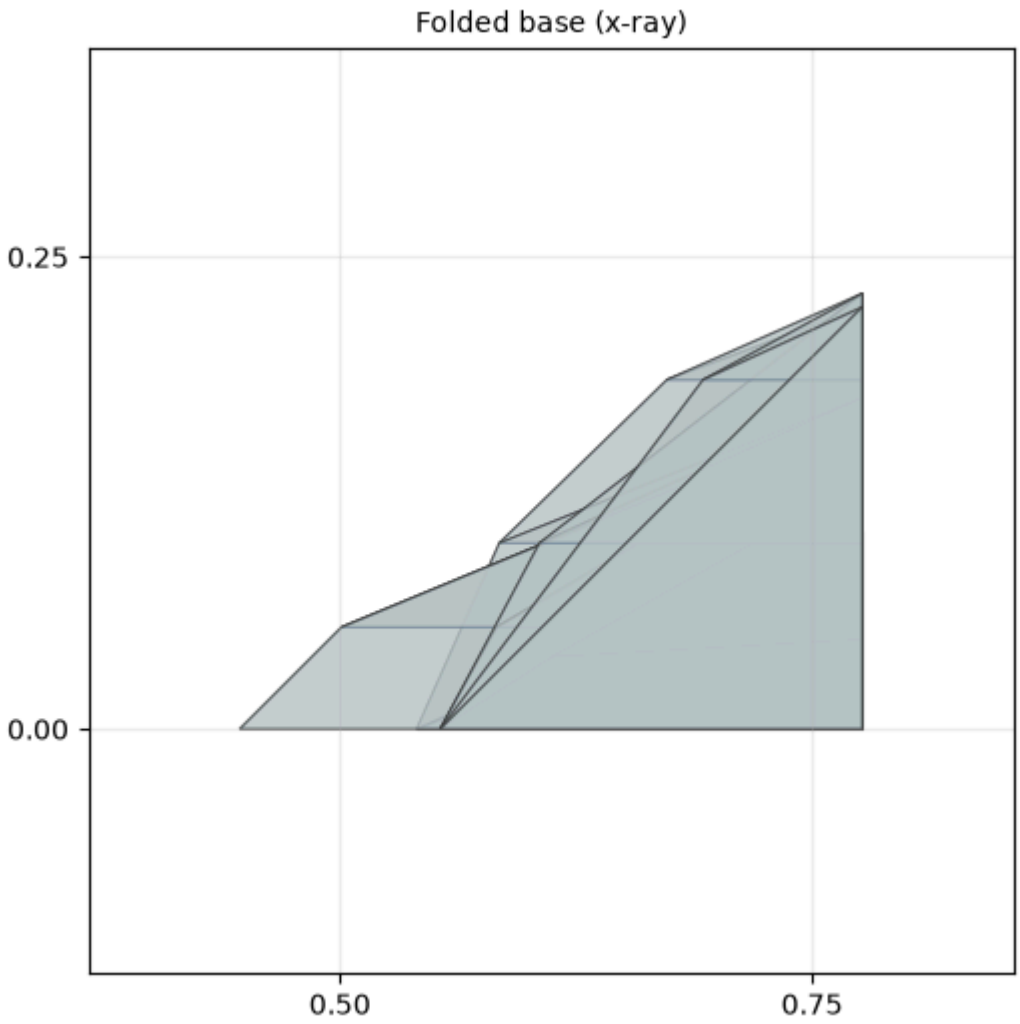

Input `folded_outline`:


```
[[[0.446872,-1.1e-05],[0.446872,-1.1e-05],[0.500762,0.053882],[0.500762,0.053882],[0.500762,0.05388
2],[0.500762,0.053882],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.05389
2],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.05389
2],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.05389
2],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.05389
2],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.05389
2],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.05389
2],[0.500757,0.053892],[0.500757,0.053892],[0.500757,0.053892],[0.500757,0.053892],[0.578937,0.08627
5],[0.578937,0.086275],[0.578937,0.086275],[0.578937,0.086275],[0.578937,0.086275],[0.578937,0.08627
5],[0.578937,0.086275],[0.578937,0.086275],[0.578937,0.086275],[0.584351,0.098459],[0.584351,0.09845
9],[0.584351,0.098459],[0.584351,0.098459],[0.584351,0.098459],[0.584351,0.098459],[0.584351,0.09845
9],[0.584351,0.098459],[0.584351,0.098459],[0.584351,0.098459],[0.584351,0.098459],[0.584351,0.098459
...
[truncated: the full input is in inputs/Q025/]
```

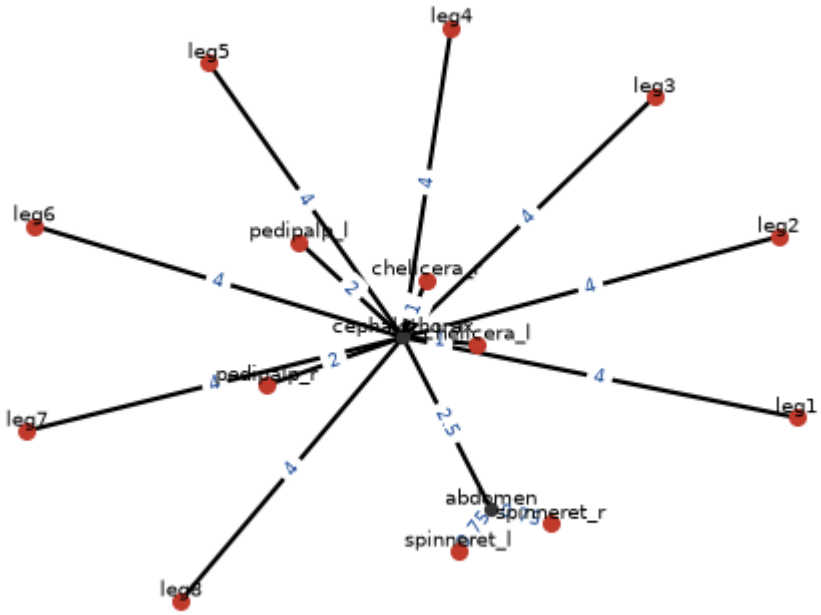

Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q026

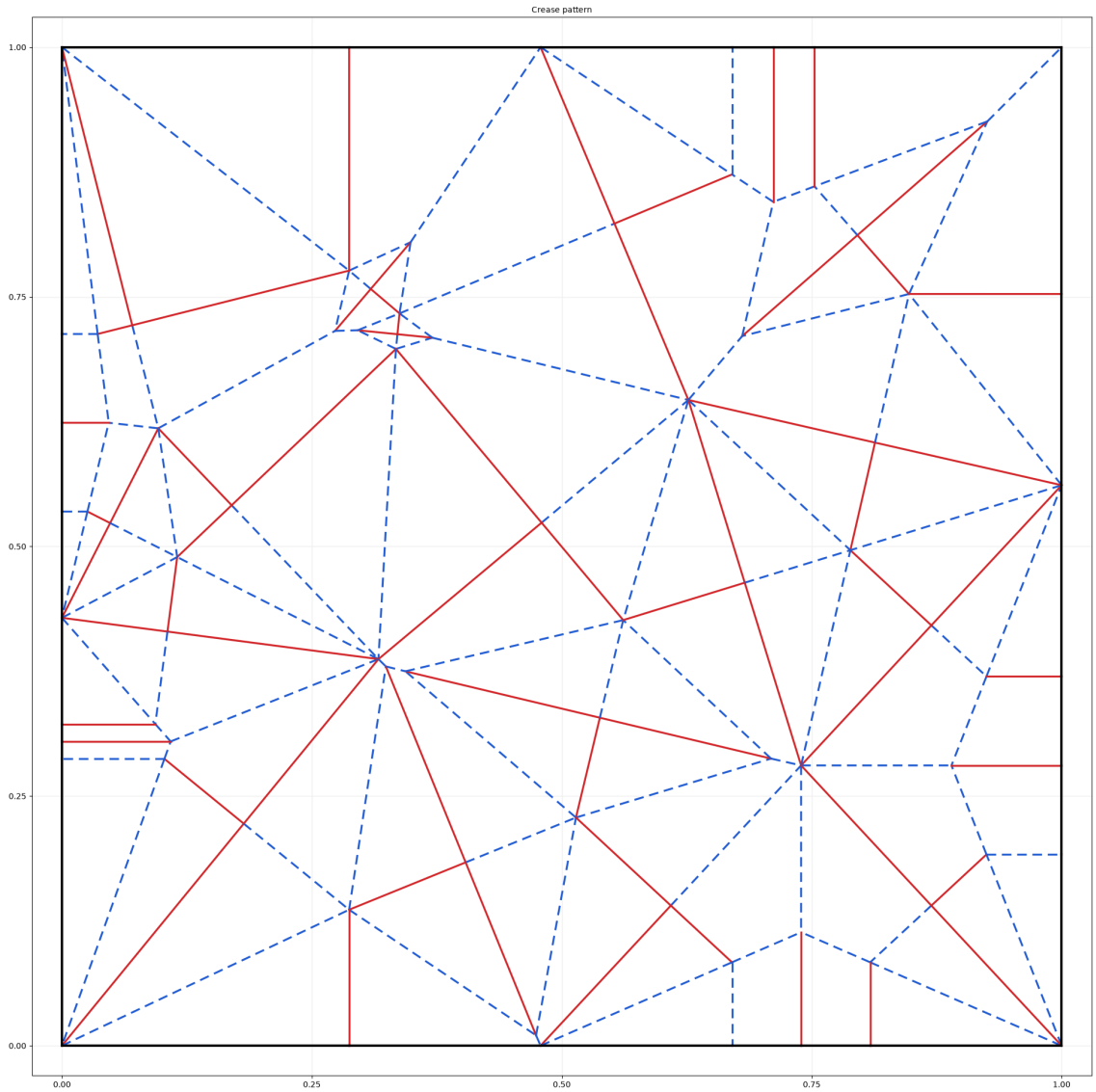
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q026/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q027

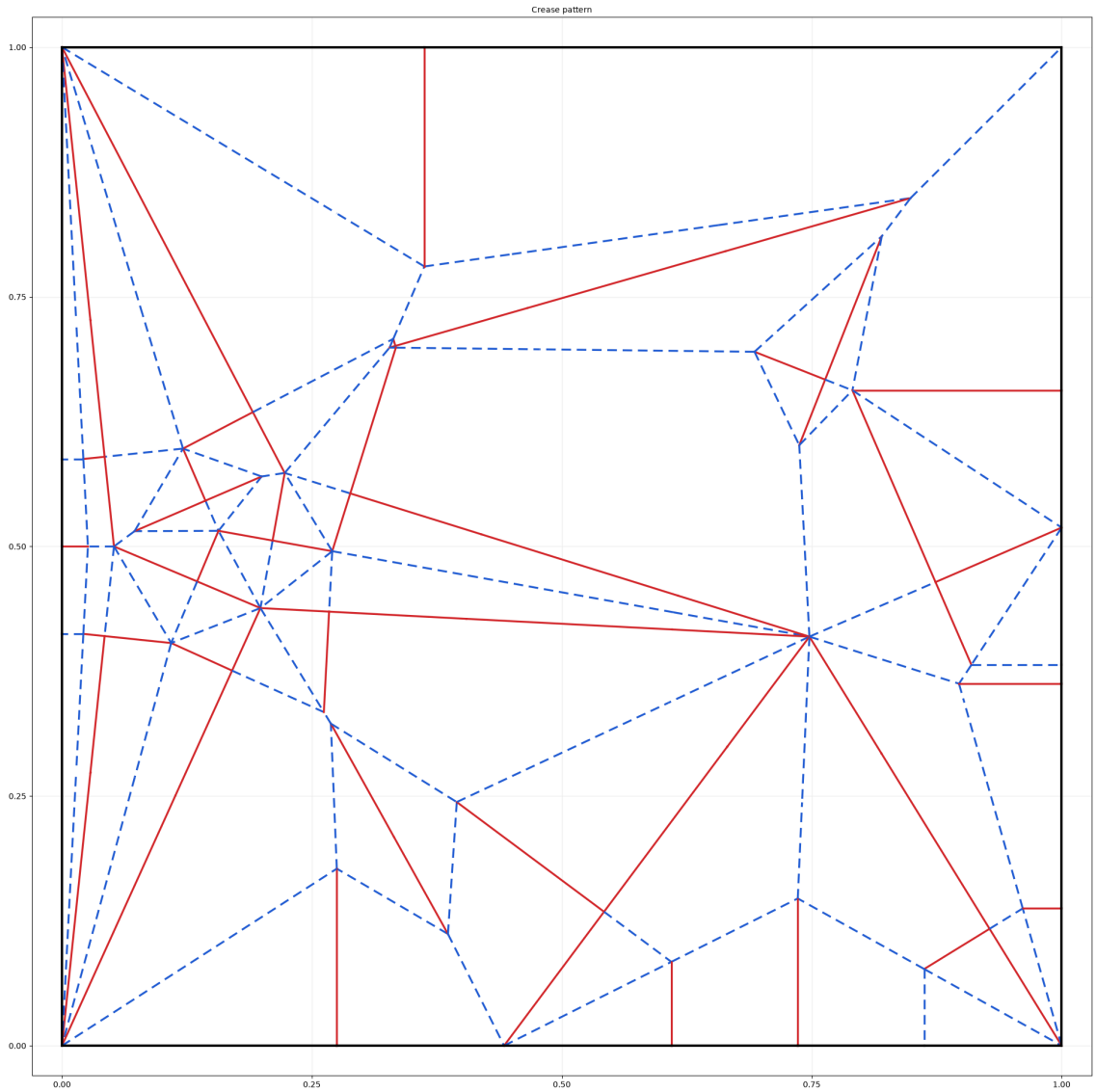
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q027/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q028

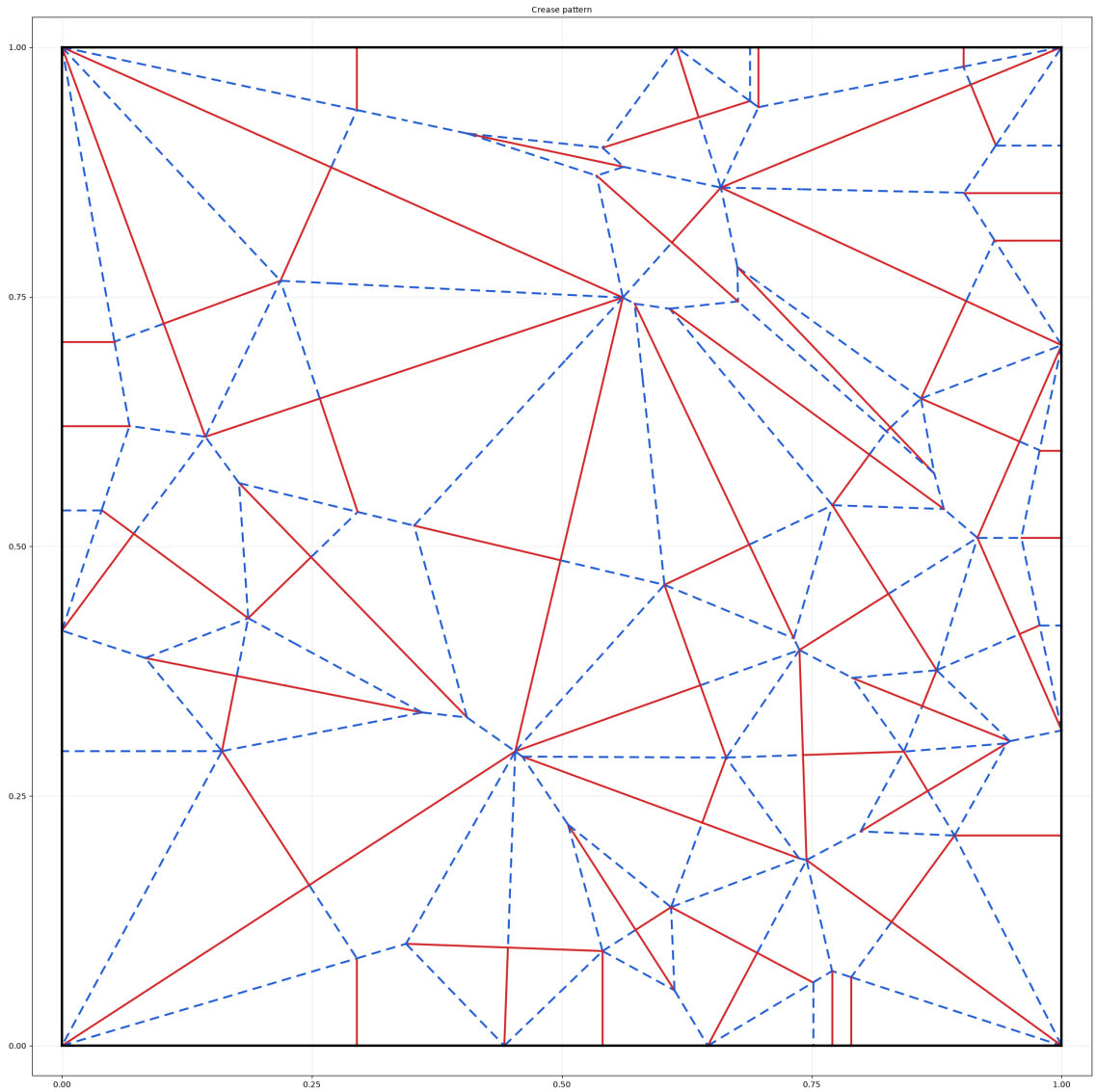
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q028/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q029

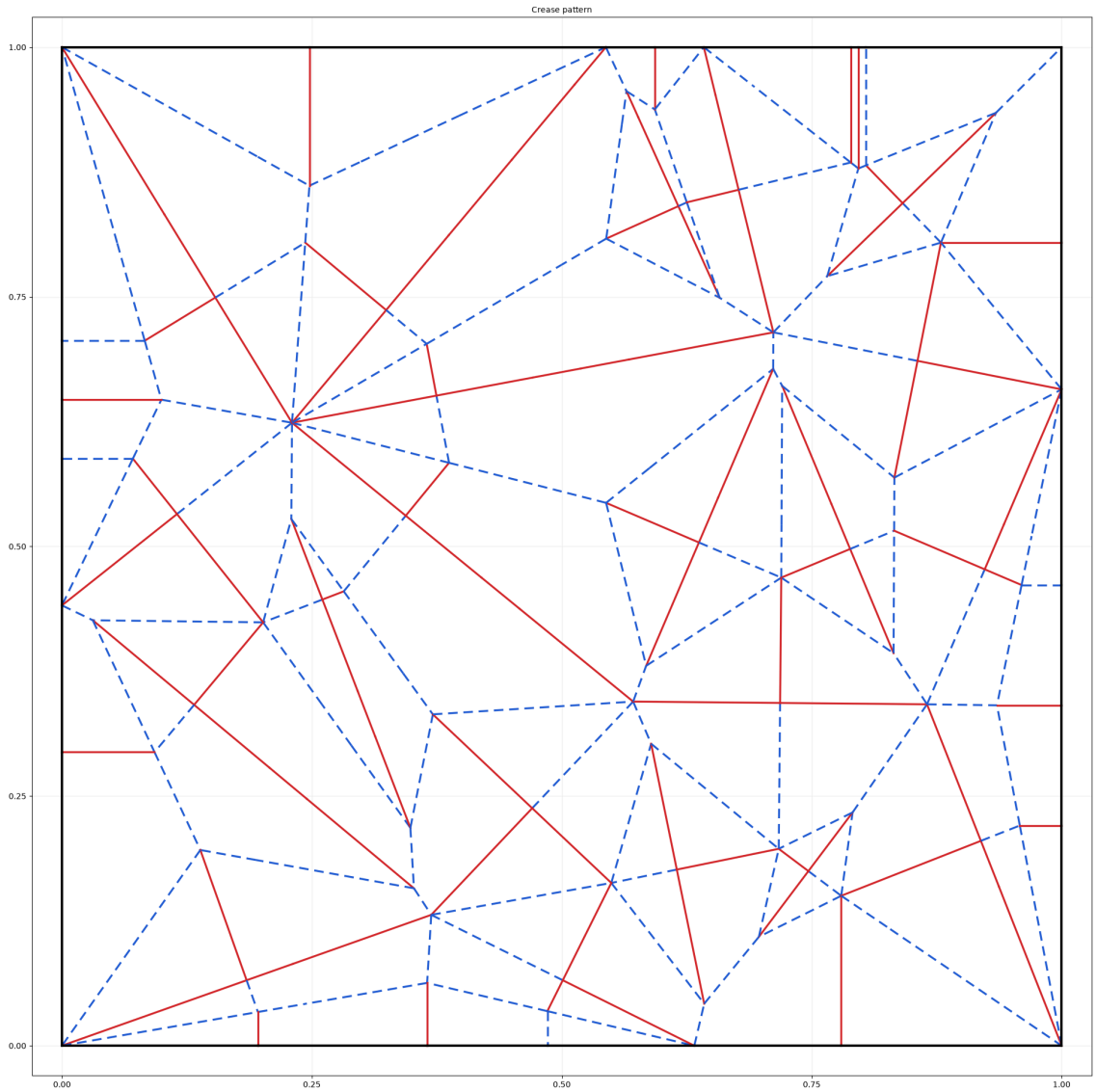
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q029/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q030

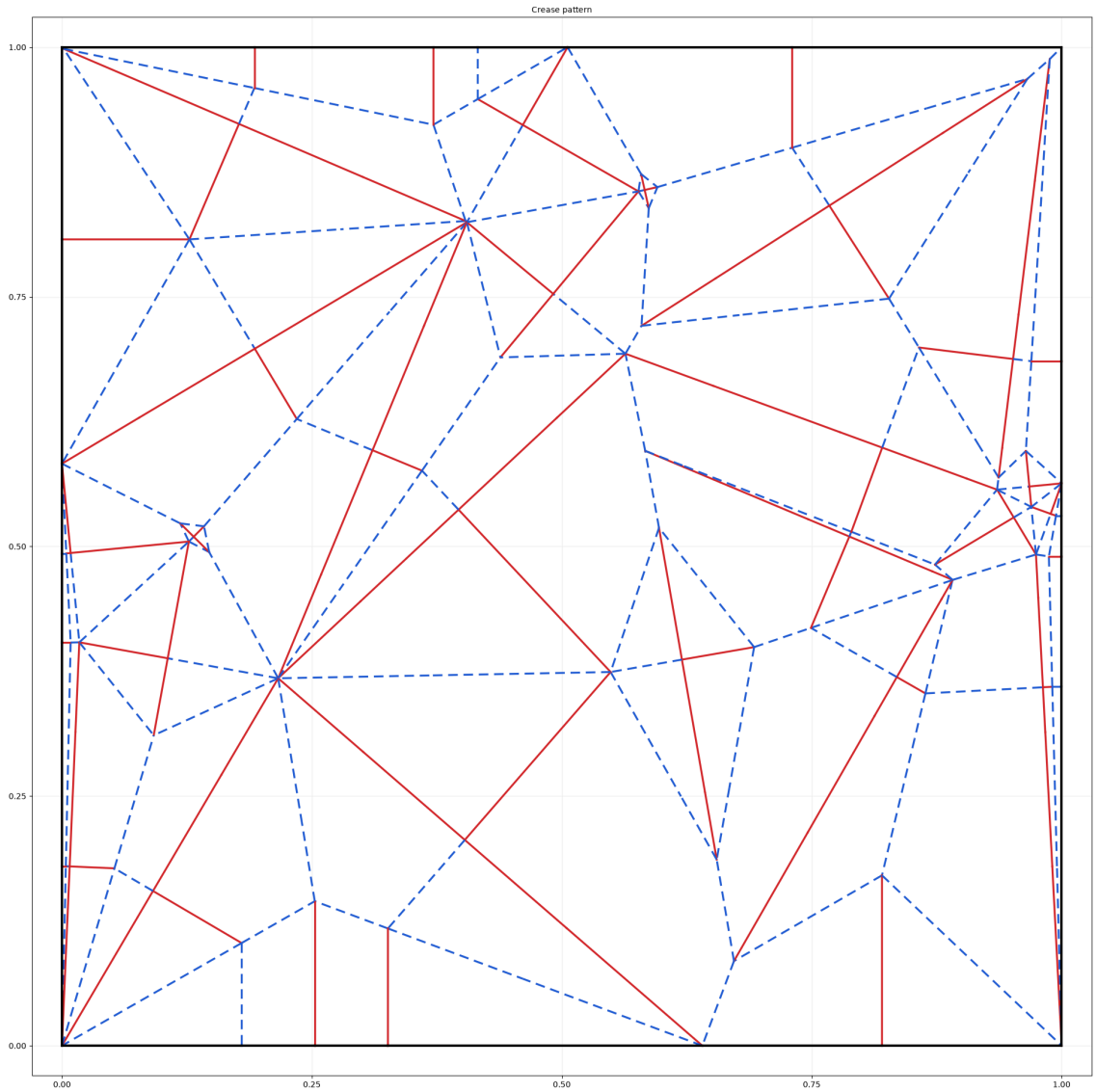
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q030/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q031

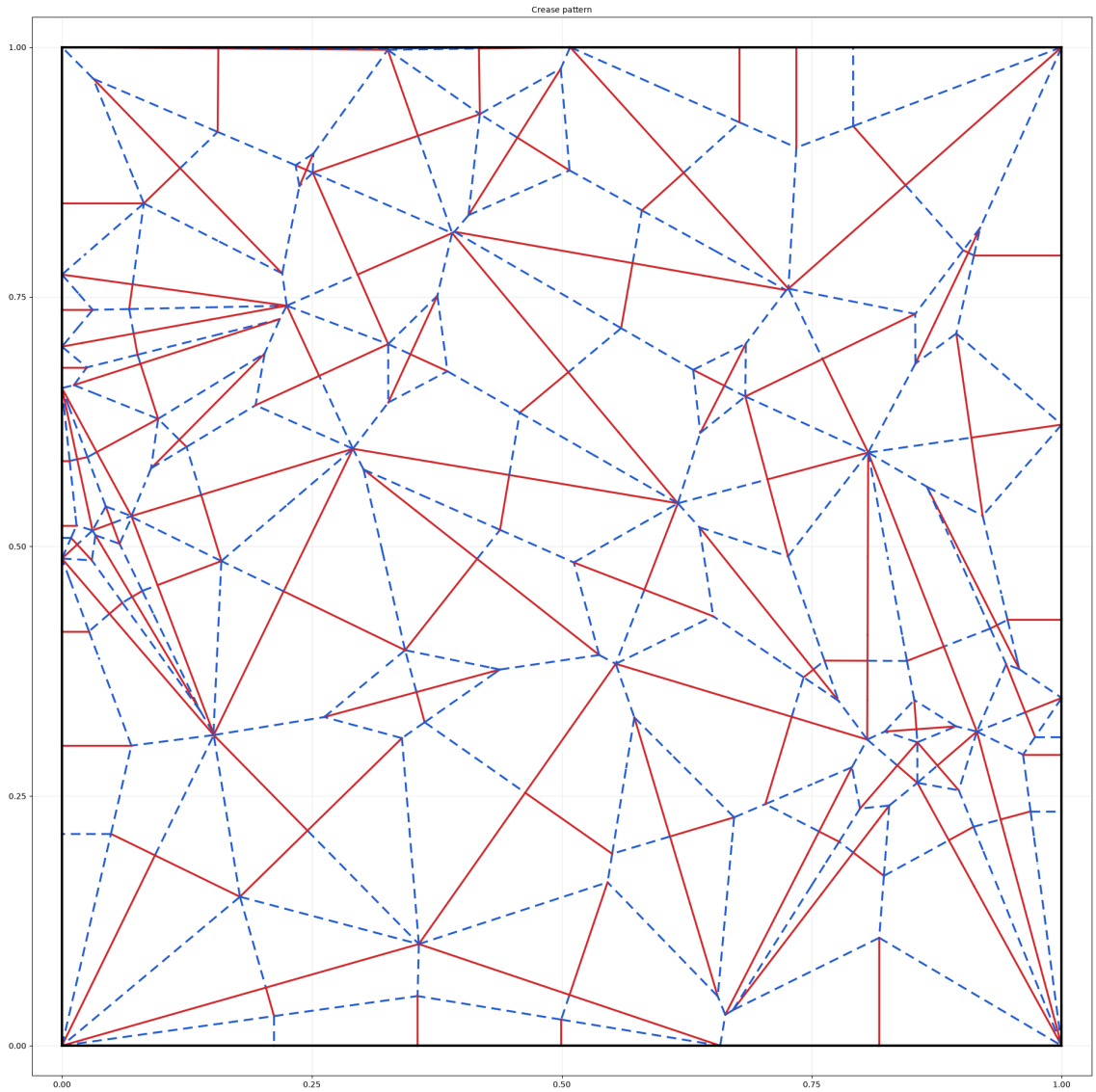
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q031/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q032

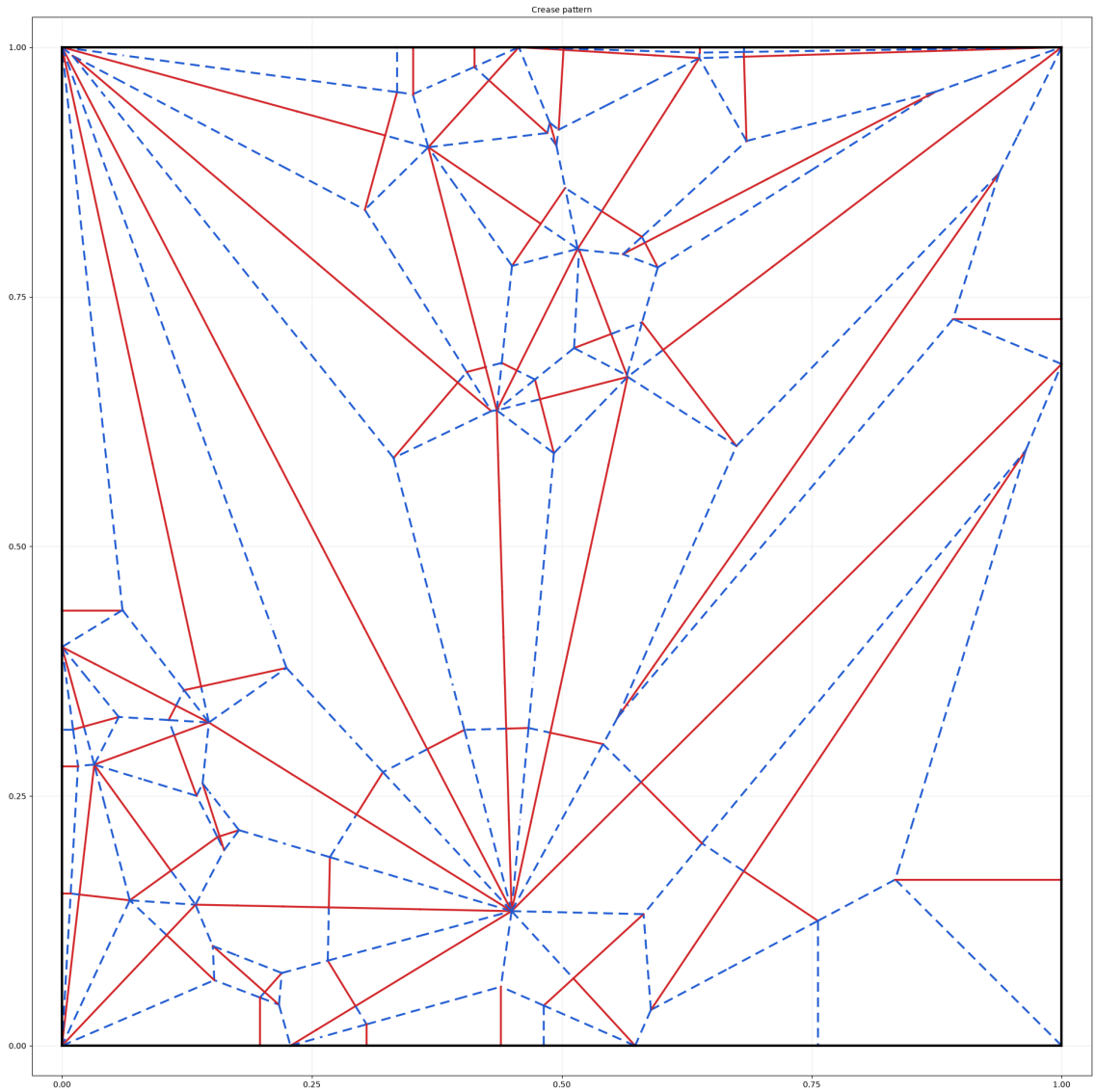
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q032/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q033

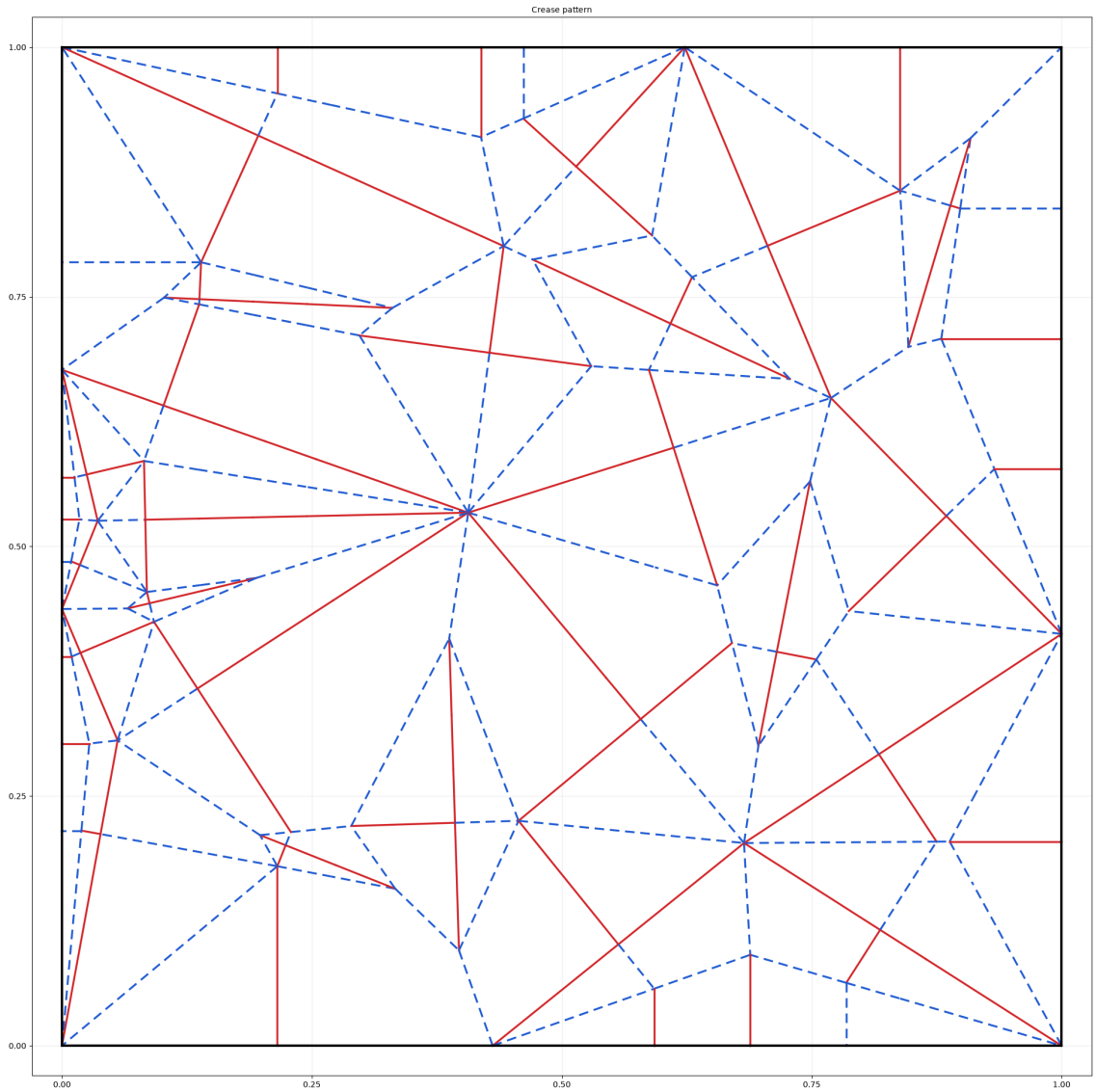
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q033/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q034

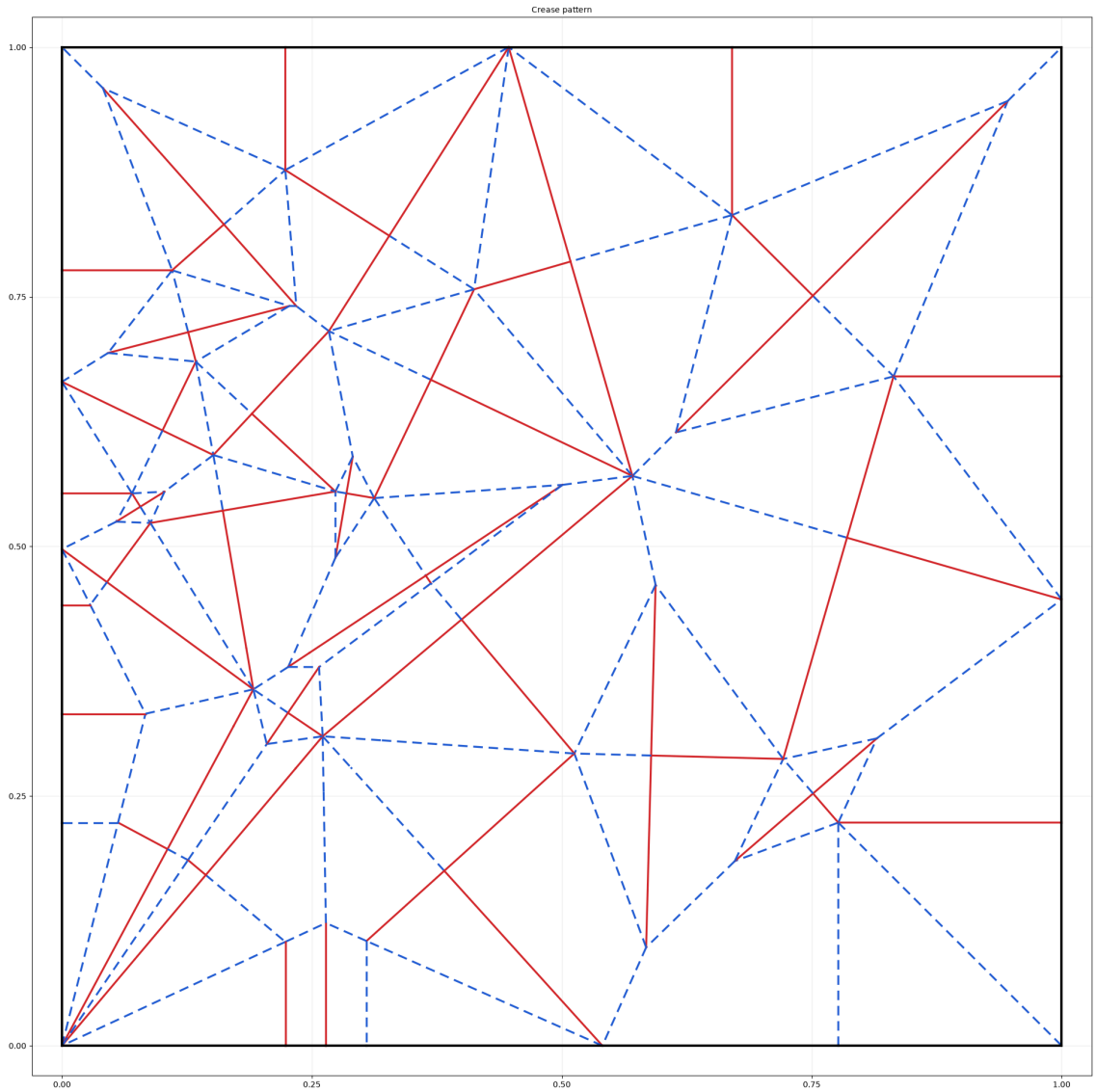
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q034/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer